mini bento

Assembly Manual

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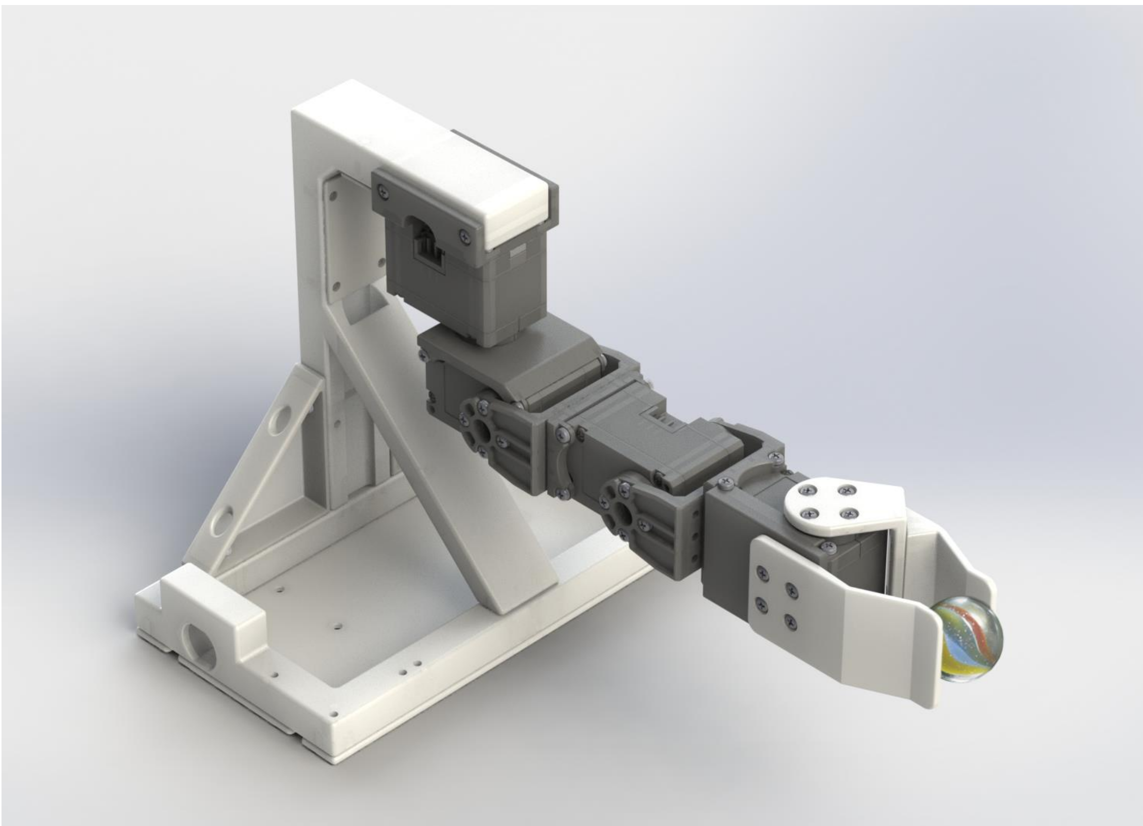


Mini Bento Arm Assembly Manual

1.1 Introduction

This guide will help you from start to finish with creating a new Mini Bento Arm including acquiring purchased parts, 3D printing custom parts, setting up the servos so that they can be controlled through some of the available software interfaces, and assembly of all parts.

1.2 Bill of materials

Please see Mini_Bento_BOM.xls for a full bill of materials for building a single Bento Arm. The bill of materials can be used to help find the parts for each of the assembly steps.

The following tools/materials are required, but not included in the bill of materials:

- RepRap 3D printer capable of printing PLA
- Precision screwdrivers (#1 Phillips and 3/32” Flathead)
- Wire cutters
- Wire strippers
- Needle nose pliers
- Gel super glue
- Label maker

Note: We have listed the distributors that we have used in the past as a point of reference, but be aware that there are multiple distributor options for each part and that we do not endorse the listed distributors.

1.3 3D Printed Parts

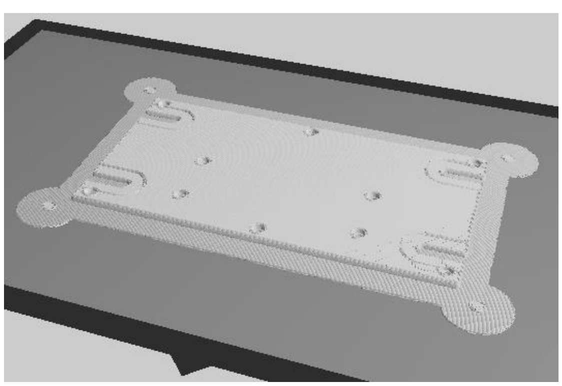
Required Tools: Makerbot Replicator 2 3D printer or similar
Required Materials: 140g of PLA filament (130g white PLA, 10g black PLA)
Estimated print time: ~9 hours

In addition to the off-the-shelf parts specified in the bill of materials, 11 additional parts need to be 3D printed for the Mini Bento Arm. In order to minimize post-processing all parts have been designed to be printed without supports in a specific orientation. The STL files for each part are located in the "3D Print Files" folder. We recommend making the following print settings into presets in your slicer to make it easier to select the recommended settings for a given part.

1. Bento_endo_raft
    a. Layer height: 0.2mm
    b. Infill pattern: linear
    c. Infill density: 50%
    d. raft margin: 6mm or greater
2. Bento_endo_noraft
    a. Layer height: 0.2mm
    b. Infill pattern: linear
    c. Infill density: 50%
    d. raft: none

NOTE1: It is very important to use the recommended print orientation since we have tried as much as possible to align the print grain in the optimal orientation for the loading of the part. You can check the orientation for each part in the screenshots below or you can download and installer [Makerware] (Version 2.4.1.24) and open the .thing files.

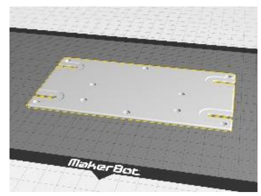
NOTE2: If you have a heated bed on your 3D printer, then you can use the "Bento_endo_noraft" settings for all parts listed below to avoid the extra step of needing to remove the rafts. If you do not have a heated bed on your printer then you should use the "Bento_endo_raft" settings on the indicated parts to avoid the issue where the corners curl up during the print. If you are finding that the corners are curling up even with the raft then you can also add "bunny_ear_rev1.STL" to the corners of your prints to help extend the raft around the corners which can also help avoid the curling issue:



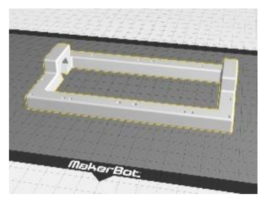
NOTE3: If you have a Makerbot Replicator 2, you can skip the slicing step and 3D print all the .x3g files that we have presliced for you. To save even more time, the items with material properties “Bento_endo_noraft” (i.e. parts 5,6,7, and parts 8-11), can be printed altogether in a given PLA color using “parts_noraft_white.x3g” and “parts_noraft_black.x3g”.

The recommended materials, print settings, and print orientations for each part are outlined below and also detailed in “Mini Bento Arm_V2_3Dprinting_guide.xlsx”. You can use this information to slice each part in the slicer for your 3D printer and then proceed to print the parts.

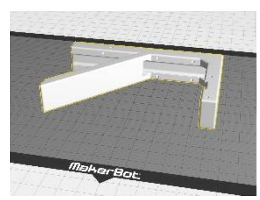
1. electronics_plate_rev2
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_raft
    c. mass: 35g
    d. time: 72 min



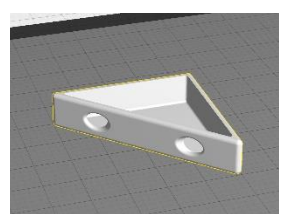
2. stand_base_rev2
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_raft
    c. mass: 42g
    d. time: 126 min



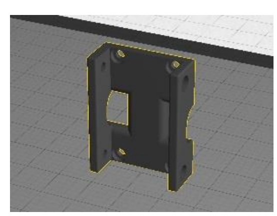
3. upper_stand_rev2
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_raft
    c. mass: 43.4g
    d. time: 150 min



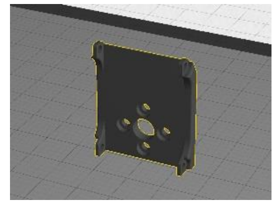
4. corner_bracket_rev2 (QTY:2)
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_raft
    c. mass: 4g per part (8g total)
    d. time: 33 min (for both parts)



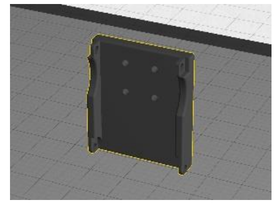
5. shoulder_adapter_rev2
    a. material: makerbot true black PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 4g
    d. time: 22min



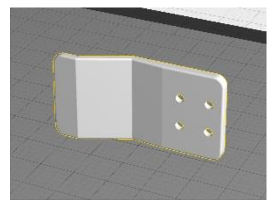
6. XL-330_side_bracket_rev2
    a. material: makerbot true black PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 3g
    d. time: 25 min



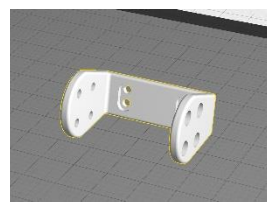
7. chopsticks_side_bracket_rev2
    a. material: makerbot true black PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 3g
    d. time: 22min



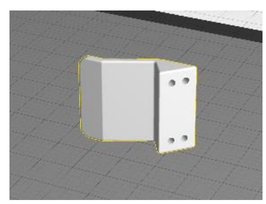
8. chopsticks_fixed_plate_rev2
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 4g
    d. time: 17 min



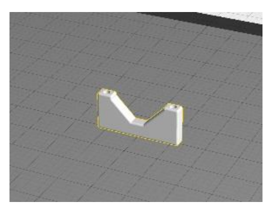
9. chopsticks_flex_bracket_rev2
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 3g
    d. time: 17 min



10. chopsticks_moving_plate_rev2
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 3g
    d. time: 17 min



11. OpenRB_150_standoff_rev2 (QTY:2)
    a. material: makerbot true white PLA or similar
    b. settings: bento_endo_noraft
    c. mass: 1g
    d. time: 10 min



Optional Parts: Also, feel free to also print QTY:1 of "cable_plate_4holes_rev2" and "cable_plate_2holes_rev2". These parts are used to help clean-up the wiring that runs from the actuators to the OpenRB-150 controller.

1.4 Dynamixel Servo Programming

Required Tools: Computer with Windows 10 or higher, [OpenRB-150] or similar, 5V/4A power supply, DC Barrel Jack (female), USB C to USB A cable
Estimated time: 0.5 hours

Table 1 – Servo settings and Joint Limits

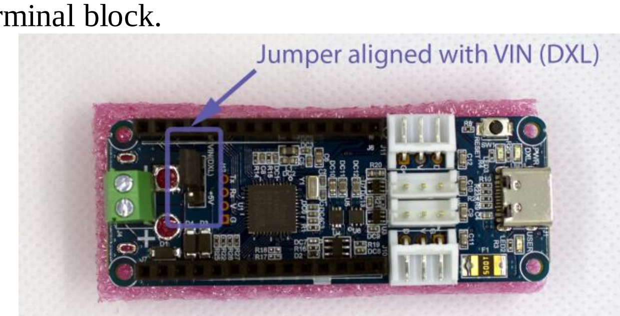
Arm Function	Servo Type	Servo ID	Baud Rate	Return Delay Time	Max Position Limit	Min Position Limit	PWM Limit
Shoulder Rotation	XL330-M288-T	1	3 (1 Mbps)	10 (20usec)	3073	1023	354
Elbow Flexion	XL330-M288-T	2	3 (1 Mbps)	10 (20usec)	2700	1150	443
Wrist Flexion	XL330-M288-T	4	3 (1 Mbps)	10 (20usec)	3248	848	354
Chopsticks Gripper	XL330-M288-T	5	3 (1 Mbps)	10 (20usec)	2658	1650	354

1. Label servos ID1, ID2, ID4, and ID5 according to the above Table 1 with a label maker

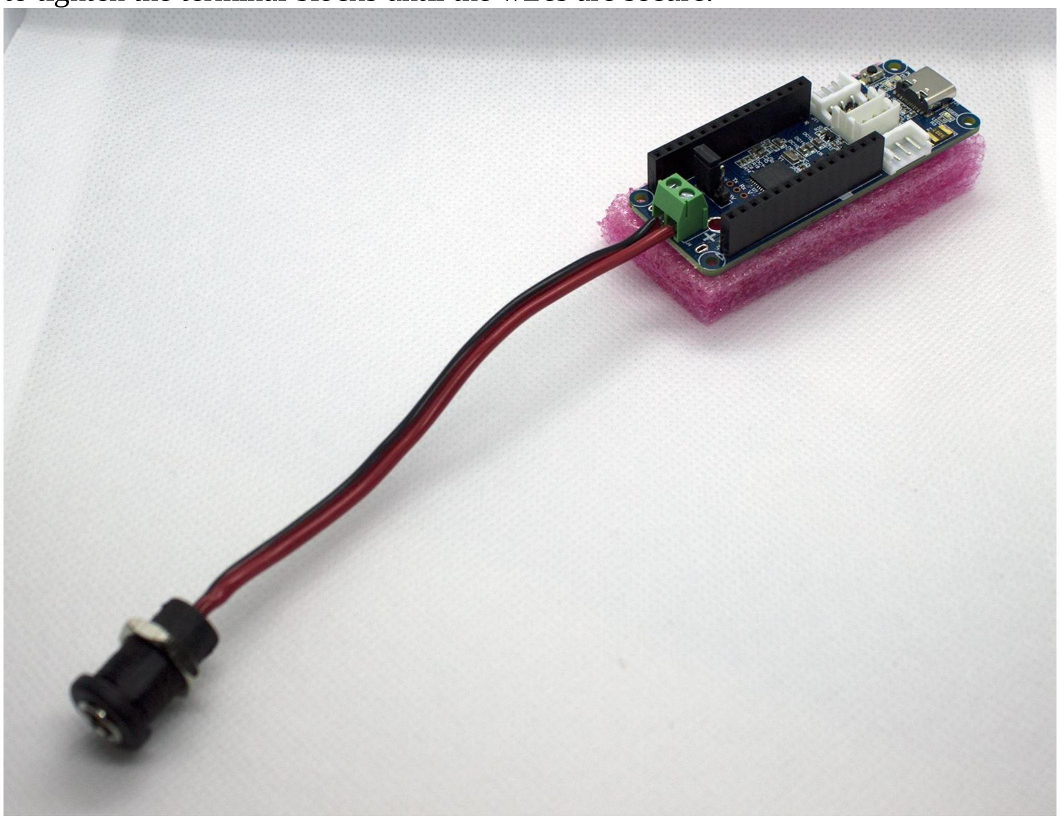
   

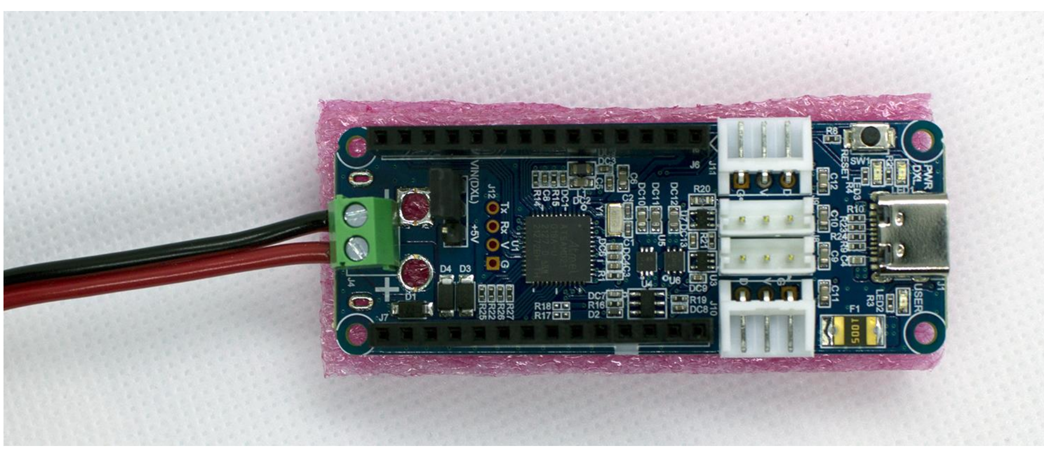
   NOTE1: Be sure to skip ID3 since it is reserved for wrist rotation which is not present in this version of the Mini Bento Arm. This will allow all the servos to match up with the naming convention in the full scale Bento Arm.

2. Ensure the power source jumper on the OpenRB-150 controller is set to VIN (DXL). When in this mode the board and connected servomotors will be powered via the green terminal block.

3. Connect the DC Barrel Jack (female) to the green terminal block on the OpenRB-150 making sure that the red wire goes into the side labelled “+” and the black wire goes into the side labelled “-“. Once the wires are in place, use a screw driver to tighten the terminal blocks until the wires are secure.

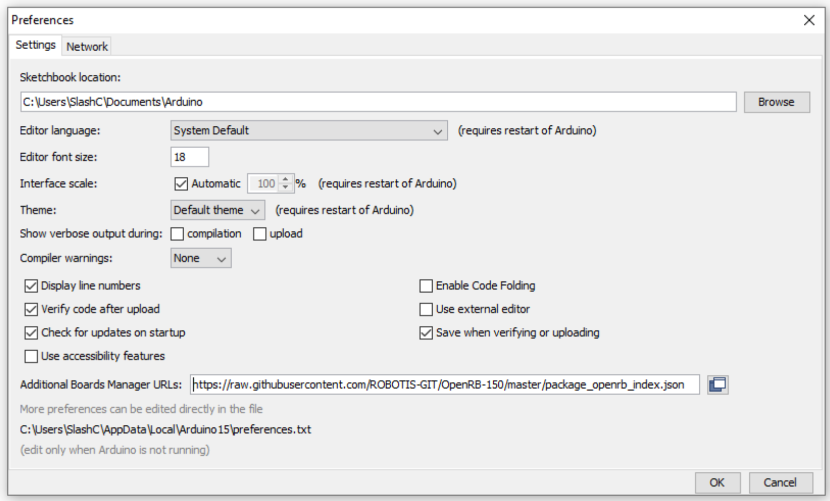




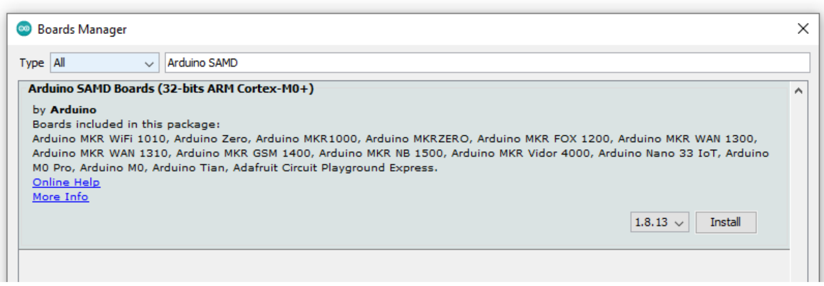
4. Install the Arduino IDE and set it up to work with the OpenRB-150
    a. Download the legacy version of the [Arduino IDE 1.8.19]() from the Arduino website and double click on “arduino-1.8.19-windows.exe” to install it.
        i. NOTE1: The newer versions of the Arduino IDE may also work, but we have not had a chance to test them yet.

b. Open the Arduino IDE and go “file > preferences”. Copy the following address into the “Additional Boards Manager URLs” textbox:

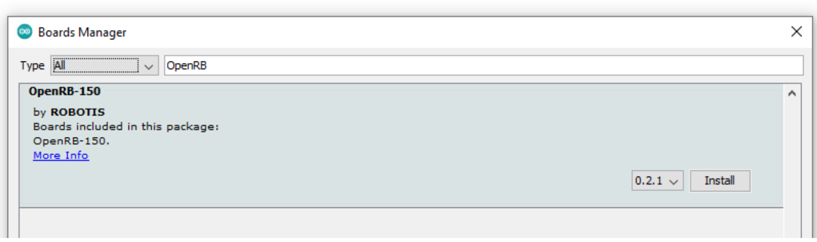
```
https://raw.githubusercontent.com/ROBOTIS-GIT/OpenRB-150/master/package_openrb_index.json
```



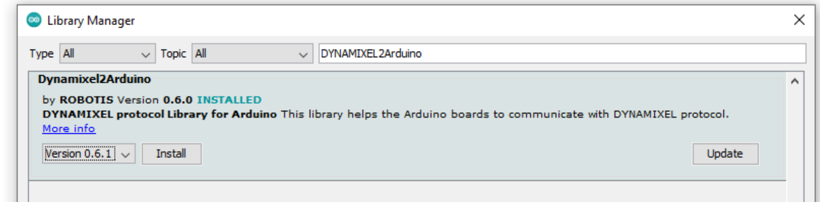
c. Install the Arduino SAMD board manager by going “tools > board > boards manager”. Search for “Arduino SAMD” and install the latest version.



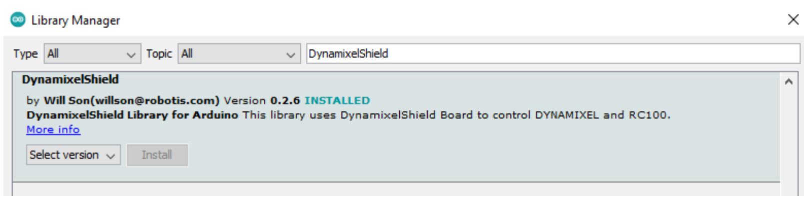
d. Install the OpenRB-150 board manager by going “tools > board > board manager”. Search for “OpenRB” and install the latest version.



e. Install the Dynamixel2Arduino library by going “sketch > include library > manage libraries”. Search for “Dynamixel2Arduino” and install the latest version.



f. Install the DynamixelShield library by going “sketch > include library > manage libraries”. Search for “DynamixelShield” and install the latest version.



5. Power on the OpenRB-150 by plugging the 5V/4A power supply into a surge protector or wall outlet and connecting the barrel plug (male) of the power supply to the DC Barrel Jack (female). The green LED on the OpenRB-150 should illuminate indicating that the board has power. A few seconds later the red LED on the OpenRB-150 should also turn on indicating that the Dynamixel bus has initialized and is also now sending power to any connected Dynamixel servos.

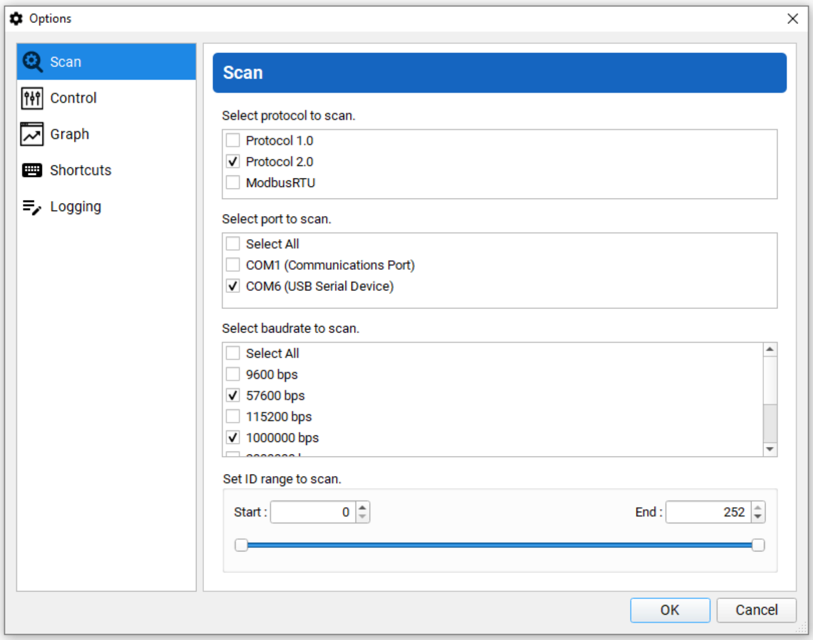
6. Connect the OpenRB-150 to the computer using the USB C to USB A cable. If this is the first time you have connected the device Windows may give some notifications that a new device is being setup.

7. Upload the “usb_to_dynamixel” sketch in the Arduino IDE
   a. Go “tools > board > OpenRB-150” and select “OpenRB-150”
   b. Go “tools > port” and select the option that has “(OpenRB-150)” next to it.
   c. Go “file > examples > OpenRB-150” and select the “usb_to_dynamixel” sketch. A new window should open in the Arduino IDE that is titled “usb_to_dynamixel”
   d. Click “upload” in the “usb_to_dynamixel” sketch to upload the sketch to the OpenRB-150

8. If it is not already installed on your computer, install the Visual C++ 2015-2019 Distributable (x86) from the following link: https://www.dropbox.com/s/3e5horhq73hm677/VC_redist.x86.exe?dl=0

9. Download and install the latest version of the “Dynamixel Wizard 2.0” Software: http://en.robotis.com/service/downloadpage.php?ca_id=10

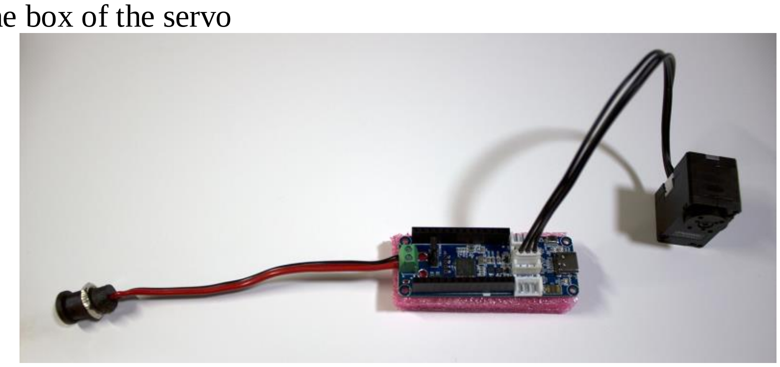
10. Once the software is installed, open the Dynamixel Wizard 2.0 and click the "Options" button. Make sure the following items are selected:
    a. Select the protocol to scan: Protocol 2.0
    b. Select the port to scan: Select the one labelled "USB Serial Device" (should be the same as step 7b)
    c. Select the baudrate to scan: 57600 bps and 1000000 bps. These are the most common baud rates that the servos are set to by default
    d. Click the "OK" button to save these options.



11. Typically the servos are initialized by default with ID=1. Since all Dynamixel servos on the bus need to have unique ID numbers you will need to repeat the following steps for each servo once when connecting to them for the first time. The parameters that will be changed for each servo are defined in the Table 1 at the beginning of this section.

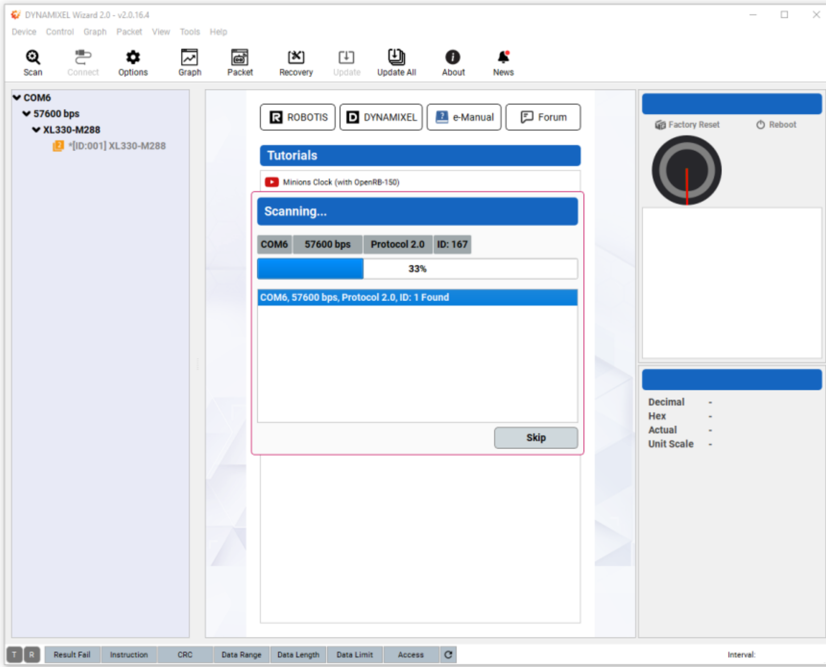
12. Disconnect the power from the OpenRB-150 by unplugging the 5V/4A power supply from the barrel connector jack (female).

13. Connect a servomotor to the OpenRB-150 using the black cable with 3 wires that came in the box of the servo

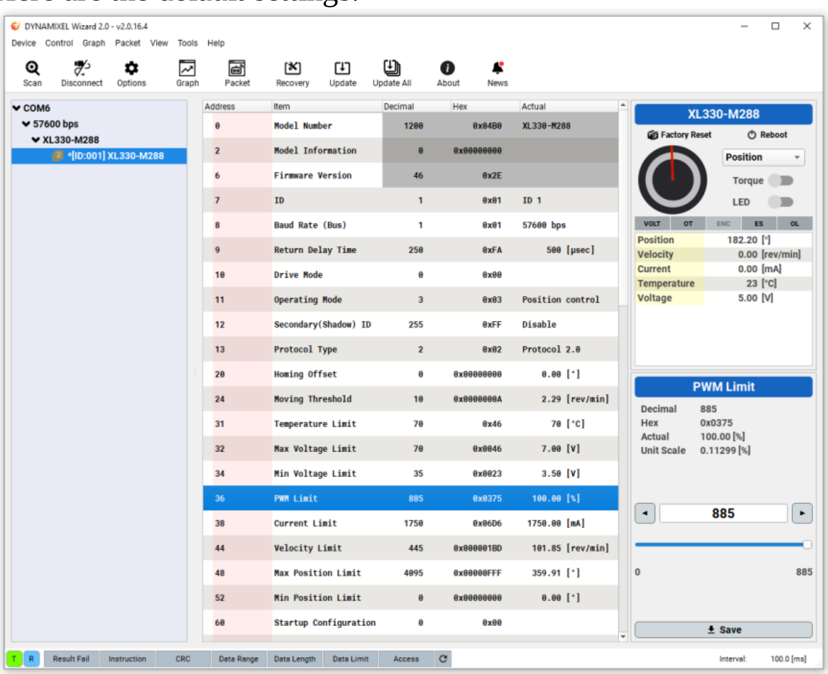


14. Reconnect the power as described in step 5.

15. Now click the “Scan” button in the Dynamizel Wizard 2.0 software. Once the servo that you connected shows up in the left pane you can click the “skip” button to stop scanning and move on to adjusting the parameters of the servo:

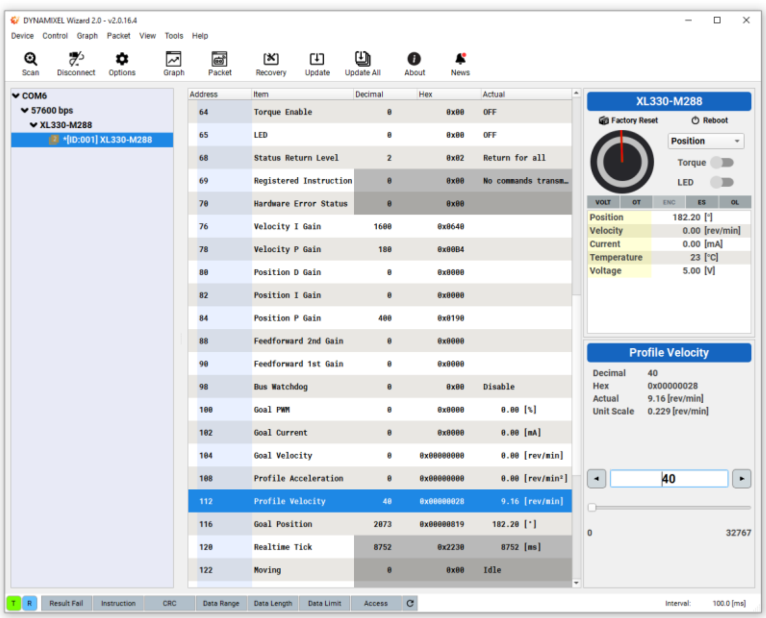


16. As mentioned previously you will need to adjust the parameters for each servo separately, but let’s setup the one for the Chopsticks gripper here as an example. Here are the default settings:

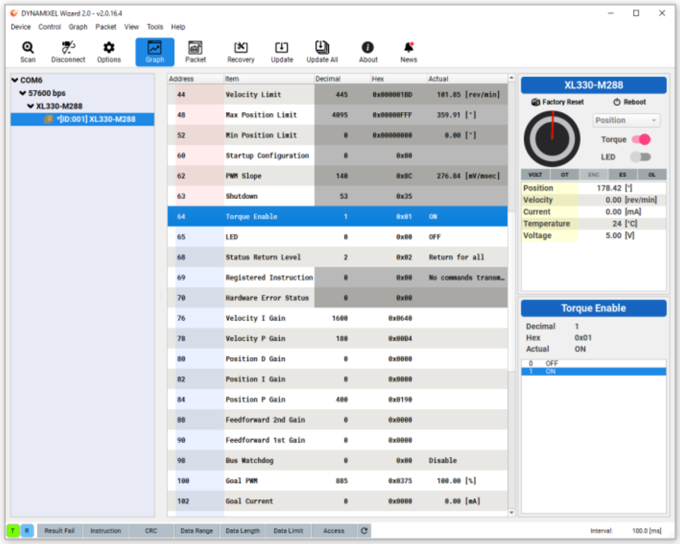


17. Change the ID to 5, baud rate to 3 (1 Mbps), and return delay time to 10 (20 usec), Min Position Limit to 1650, and Max Position Limit to 2658, and PWM Limit to 354. You will need to select the register to change the value and then click the 'save' button to make these changes.
    a. **NOTE1**: It is very important to set the max and minimum position limits before you ever connect the servos to another software in order to prevent the servos, brackets, or cables from being damaged by moving beyond their range of motion. You can check the range of motion of a given joint or mechanism by manually moving the joint and reading out the max and min positions from the 'present position' register when the torque is turned off.
    b. **NOTE2**: The PWM limits are also an effective way of helping prevent the arm from damaging itself or overloading by pushing too hard into the table or other obstacles. If you want to carry heavier payloads you can increase the elbow and wrist flexion servos to as high as 354 without them overloading, but we think these lower values should work well for most applications. After you change the value in the register it will not be active until you cycle the power on the servo.
    c. **NOTE3**: Be very careful when setting the baud rate and ensure that you do not accidentally set it to higher than 1 Mbps as this will make it so that the actuator can no longer communicate with the OpenRB-150 and a separate U2D2 will be required to set the baud rate back to a lower value.

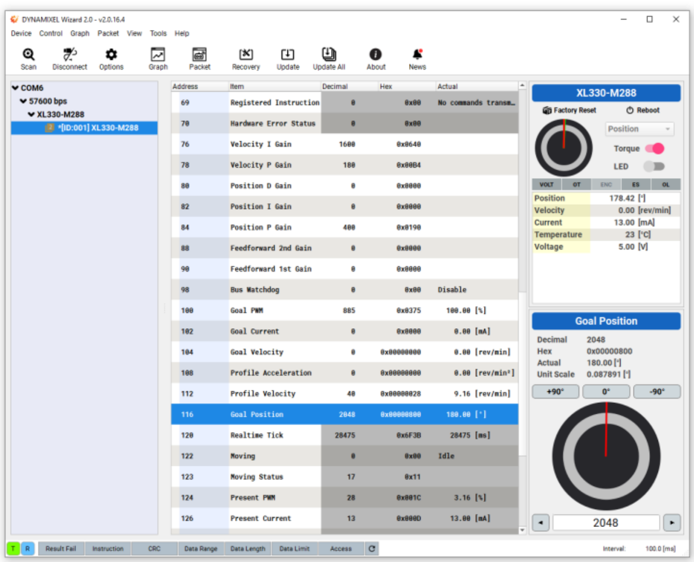
18. You can also try moving the servo in the Dynamixel Wizard 2.0 software
    a. Set the "Profile Velocity" register (address 112) to a reasonably low testing speed such as 40 (9 RPM)



b. Set the “Torque Enable” register (address 64) to 1 to turn on torque to the motor.



c. After clicking on the “Goal Position” register (address 116) you can then change the goal position using the rotational control that is built into the software. When you are done moving the servo please re-center it by setting the goal position to 0 degrees. This will ensure the servo horn is aligned properly for assembly. NOTE: It is very important to set the moving speed first otherwise the servo defaults to moving as fast as possible which will increase the chances of causing damage if the settings are incorrect.



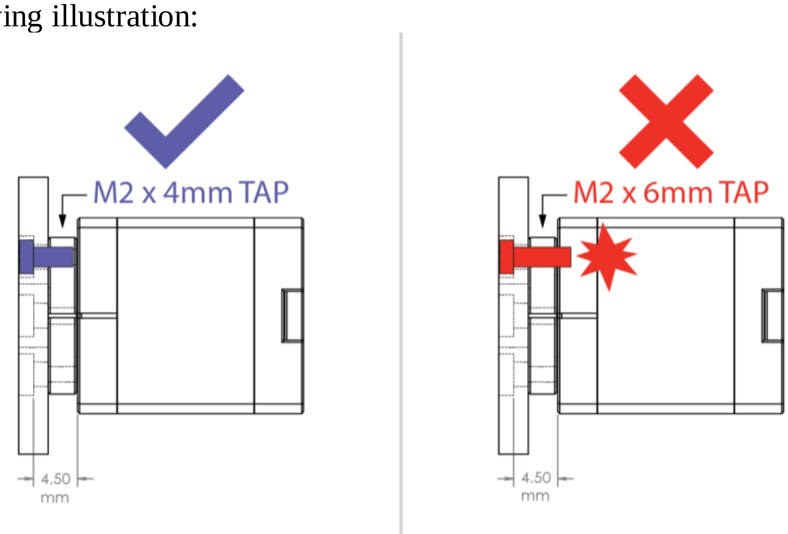
19. Click the “Disconnect” button in the Dynamixel Wizard 2.0 software and then repeat steps 11 to 17 to program the next servo and set its parameters according to Table 1 until you have finished with all the servos.

1.5 Arm Assembly

Required Tools: #1 Drive Phillips Screwdriver (3.8mm diameter or less), needle nose pliers, wire cutters, wire strippers, soldering iron (optional)
Required Materials:
Estimated time: 1.5 hours

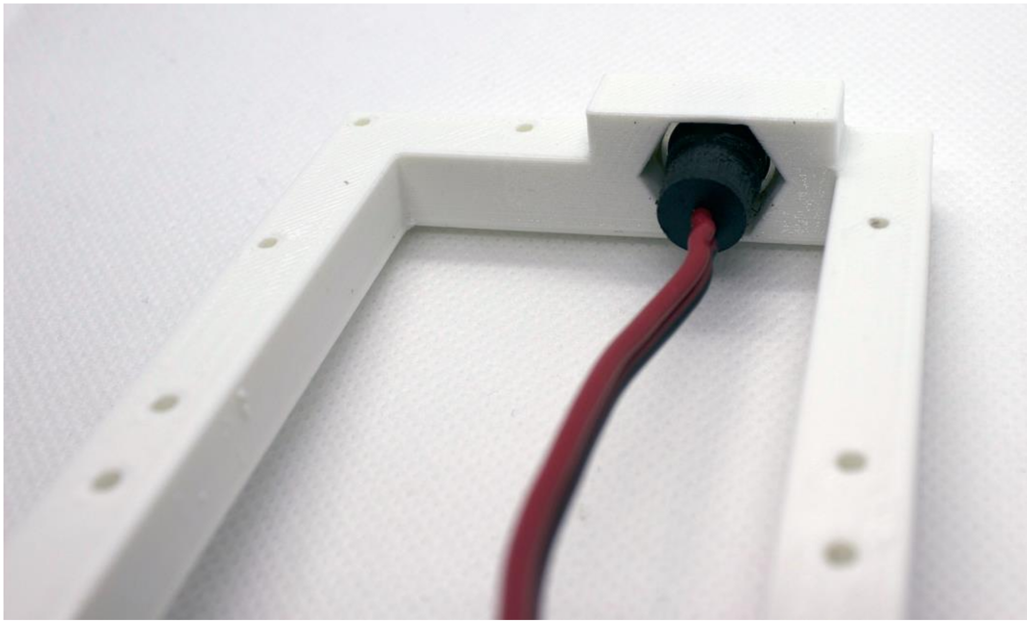
When going through each of the following steps it may help to open the 3D model in a 3D CAD program (Solidworks or STEP) to find the names of parts and their orientations that they need to be assembled in. You can also see some of the names and part orientations in the Mini Bento Part Orientation Guide.

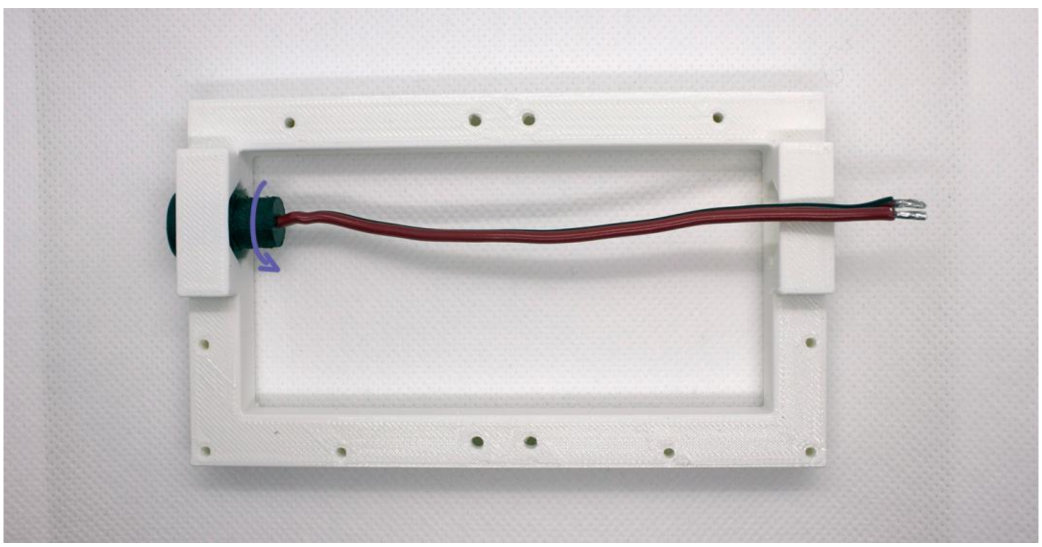
1. To make the next assembly steps easier we recommend organizing the screws that came with the servomotors and brackets into 4 separate containers or bags. If you do not have any containers then you can organize them into the small ziplock bags that came with the XL-330 actuators.
    a. M2x8mm TAP screws (black)
    b. M2x6mm TAP screws (black)
    c. M2x4mm TAP screws (black)
    d. M2x4mm screws (silver) and M2 nuts (silver)
    e. NOTE: Be careful to use the correct screw length and type for each step otherwise parts may be damaged. For example, if you use 8mm screws in the servo horns it may permanently damage the servo as seen in the following illustration:



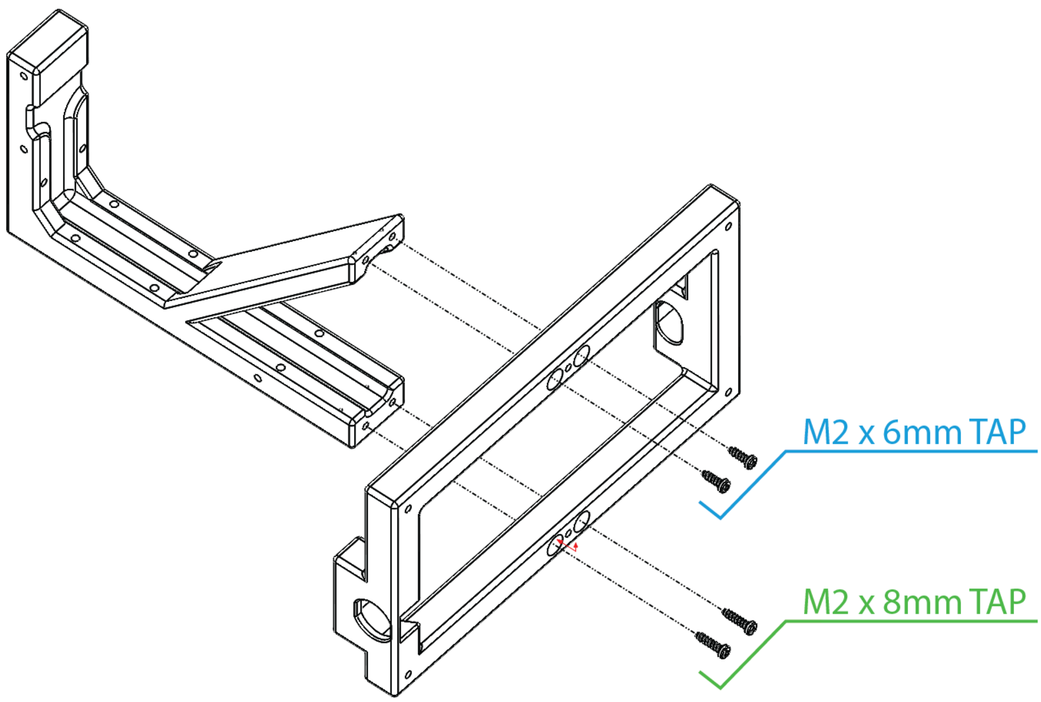
2. Carefully unscrew the hex nut from DC Barrel Jack (female). The plastic threads are very soft and prone to cross-threading. If the nut starts to bind back off and try again. If after a few tries you are not able to get the nut off without binding then the connector may have already have been cross-threaded at the factory. In that case you can use needle nose pliers to unscrew the hex nut.

3. Connect the DC Barrel Jack (female) to stand_base by feeding the wires through the hole in the outside of the stand. Once the connector is flush with the outside of the stand you can slide the hex nut onto the wires and into the hex shaped hole on the inside of the stand. You can then rotate the part of the connector on the inside of the stand to get the nut to tighten using needle nose pliers.

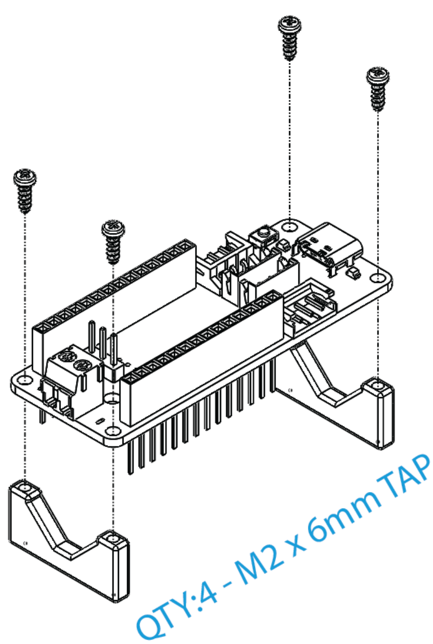




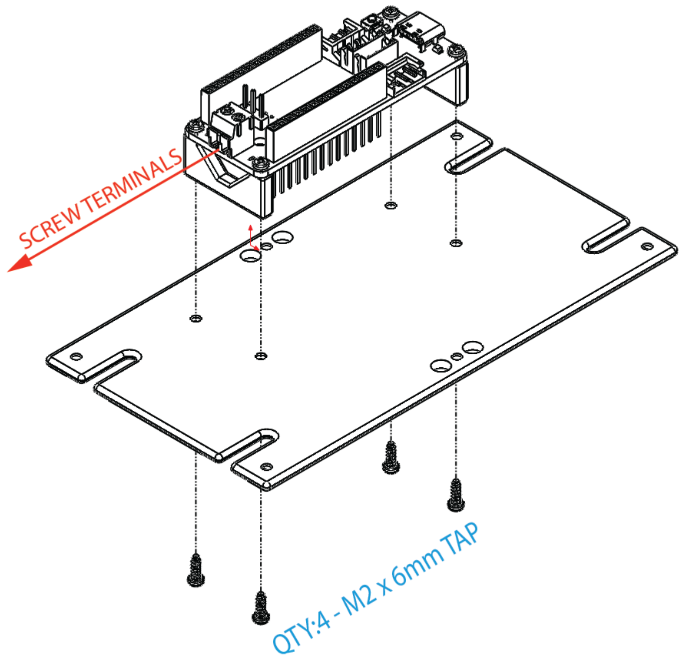
4. Attach stand_base to upper_stand using QTY:2 M2x6mm TAP screws in the holes on the front (45 deg section) of the stand and using QTY:2 M2x8mm TAP screws in the holes on the back of the stand.



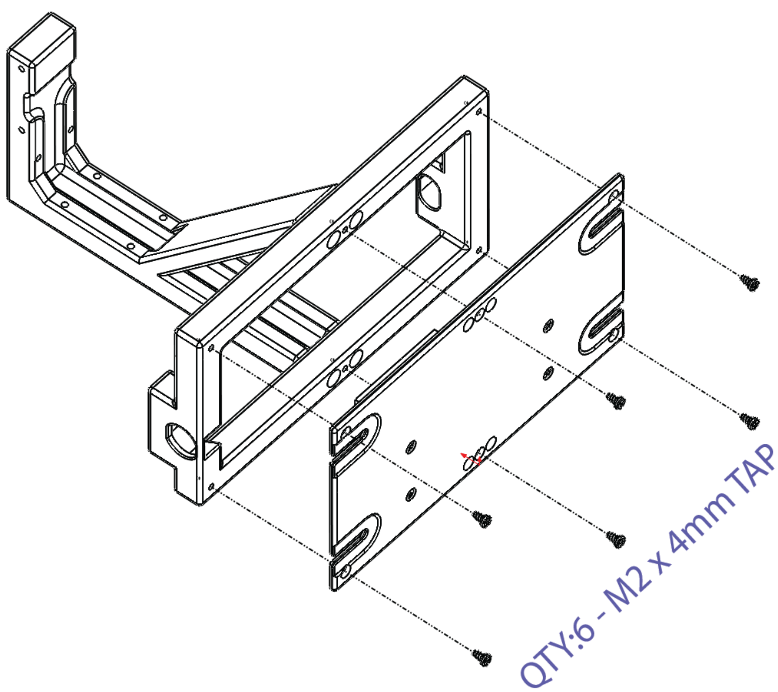
5. Attach QTY:2 of OpenRB_150_standoff to OpenRB-150 using QTY:4 M2x6mm TAP screws.



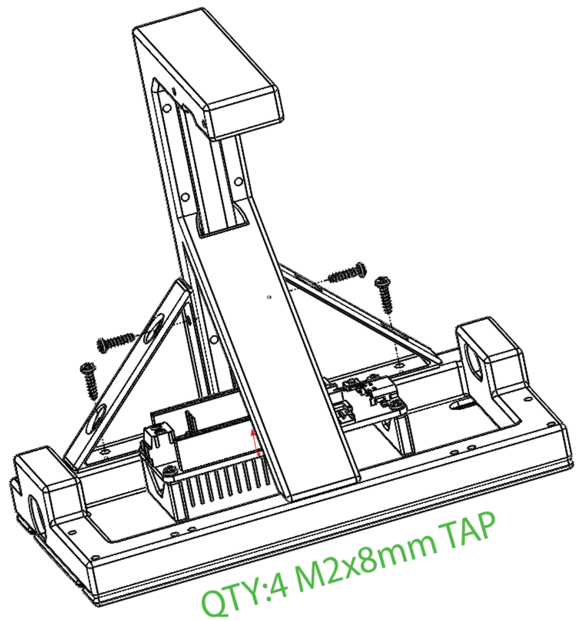
6. Attach the OpenRB_150 with the installed standoffs to the electronics_plate using QTY:4 M2x6mm TAP screws. The side of the OpenRB_150 with the screw terminals should face towards the left as shown below:



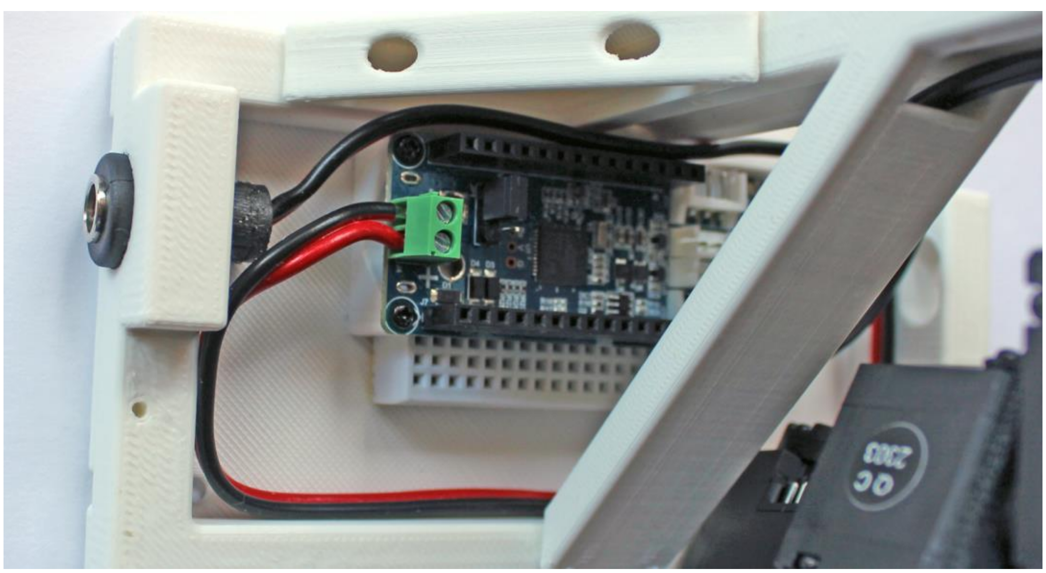
7. Attach electronics_plate to stand_base using QTY:6 M2x4mm TAP screws from the FPX330-H101 bracket kit.



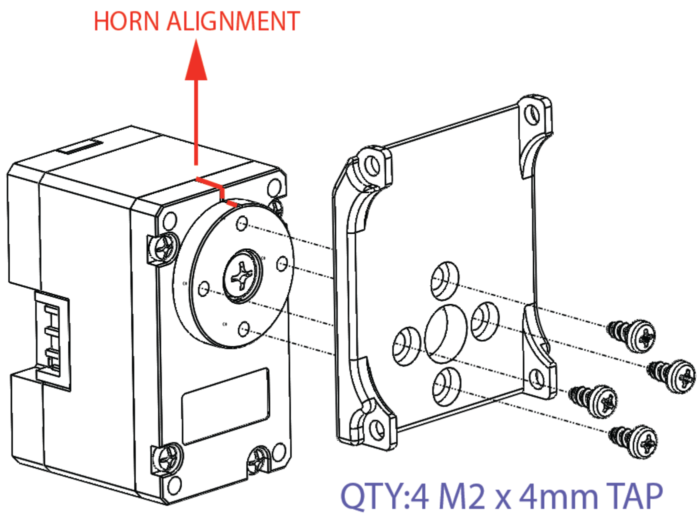
8. Attach QTY:2 corner brackets to stand assembly using QTY:4 M2x8mm TAP screws.



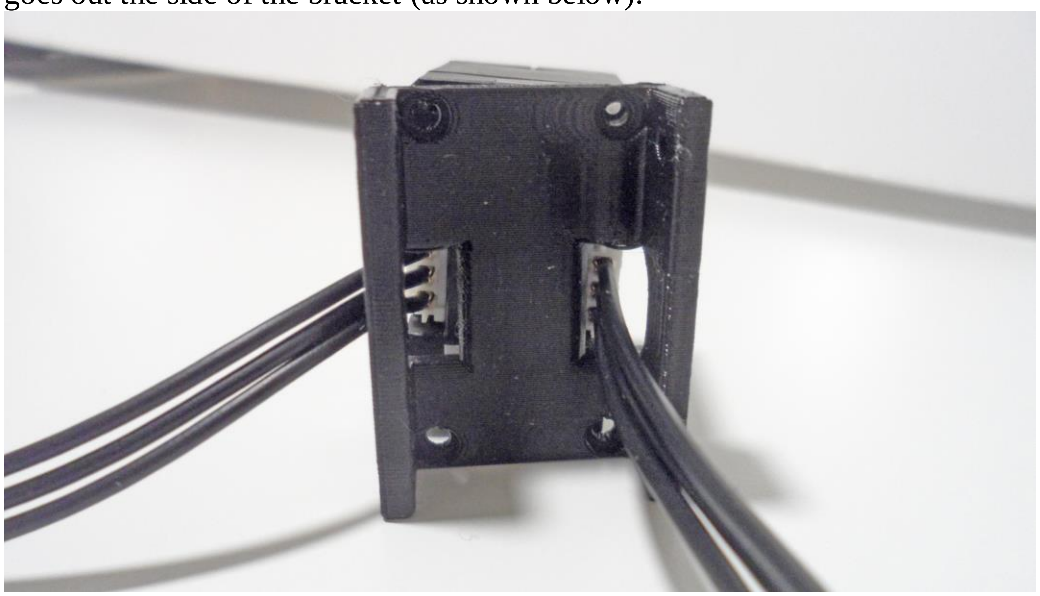
9. Attach the power wires from the DC Barrel Jack (female) to the OpenRB-150 making sure that the red wire goes into the side labelled "+" and the black wire goes into the side labelled "-". The wires fit nicely into the base when wrapped between the OpenRB-150 and the upper stand and then back around the base to the wire terminals. Once the wires are in place, use a screw driver to tighten the terminal blocks until the wires are secure.



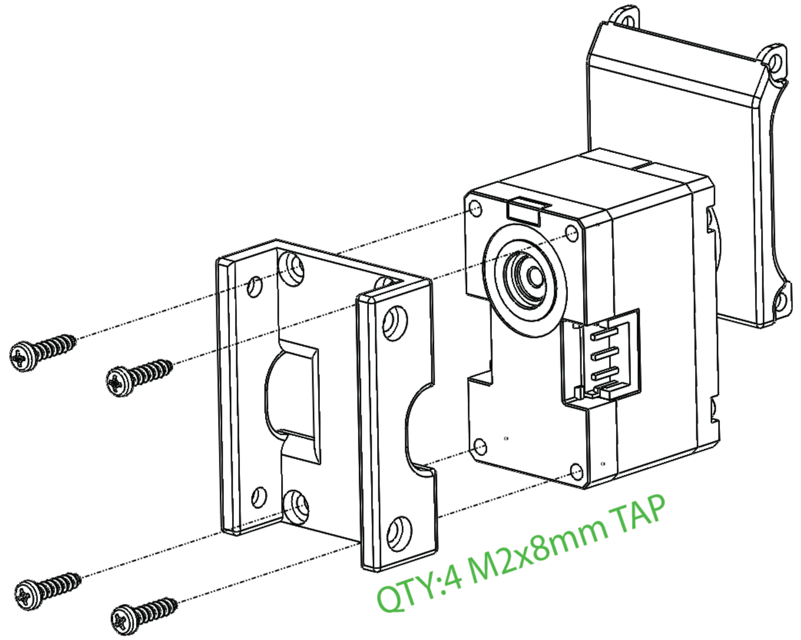
10. Attach XL-330_side_bracket to the shoulder actuator using QTY:4 M2x4mm TAP screws. Ensure that the notch on the servo horn aligns with the notch on the servo casing and that the bracket is also aligned as shown below:



11. Attach QTY:2 180mm cables to the shoulder actuator (XL330-M288-T, ID:1). Place the shoulder_adapter against the back of the actuator making sure that the right cable feeds through the slot in the bracket and the left cable folds over and goes out the side of the bracket (as shown below).

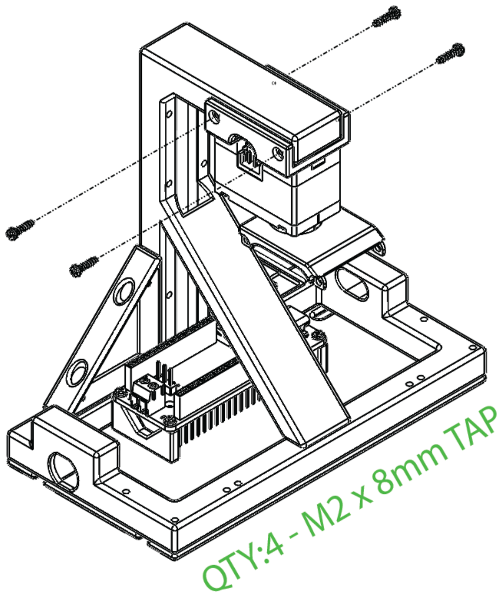


12. Attach shoulder_adapter to the shoulder actuator using QTY:4 M2x8mm TAP screws.

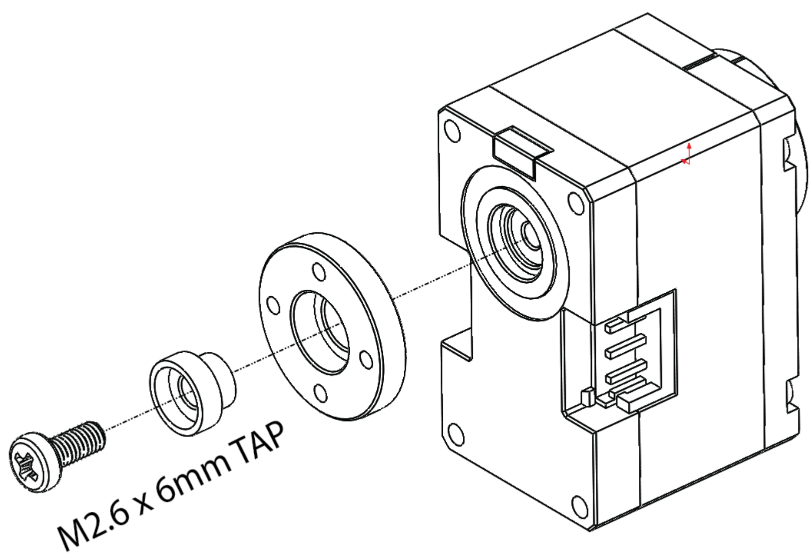


13. Carefully guide the cable on the right side (when viewing the stand from behind) into the cable channel of the upper_stand, so it lies flat and the cable on the left side folds down and out of the shoulder_adapter (see photo below).

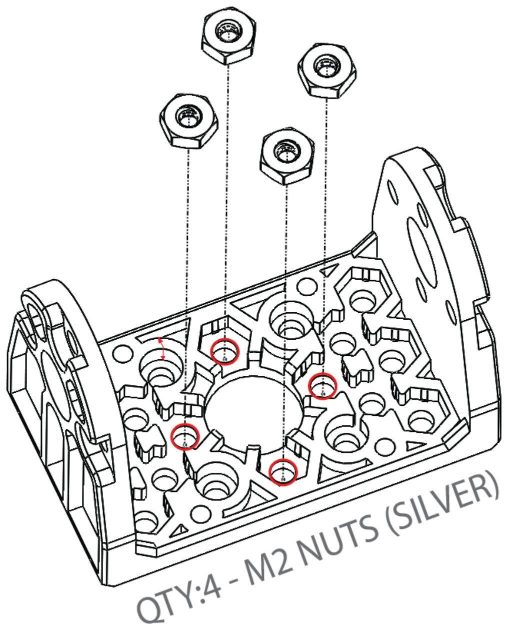
14. Attach the shoulder_adapter to the upper_stand using QTY:4 M2x8mm TAP screws. Once attached you can tuck the cable in the channel through the slot in the 45 degree support to help keep it out of the way.



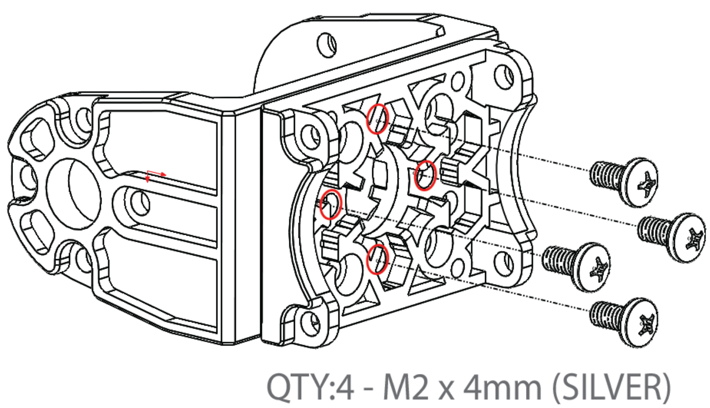
15. Attach XL-330_idle_bearing and XL-330_idle_horn to elbow actuator (XL330-M288-T, ID:2) using QTY:1 M2.6x6mm TAP screw.



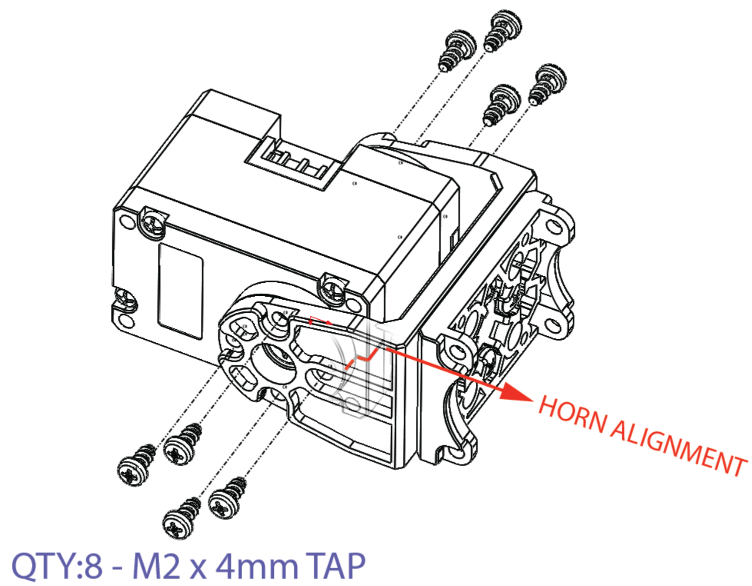
16. Insert QTY:4 M2 nuts (silver colored) into the 4 closest holes surrounding the central hole in the FPX330-H101 elbow flexion bracket.



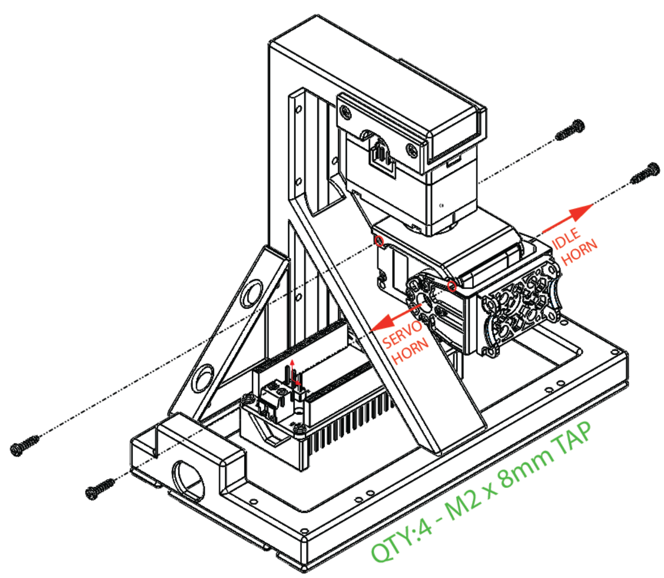
17. Attach the FPX330-S102 bottom bracket to the elbow flexion bracket using QTY:4 M2x4mm screws (silver colored).



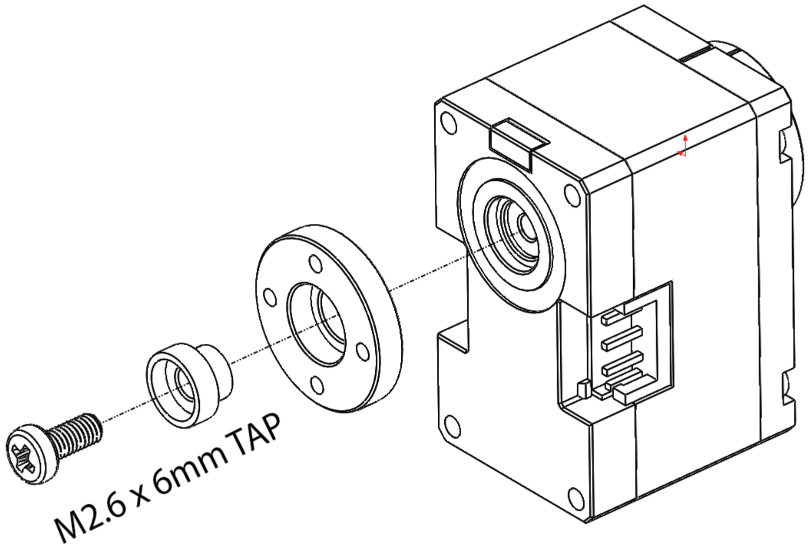
18. Attach the elbow flexion bracket to the elbow actuator using QTY:8 M2x4mm TAP screws. Ensure that the notch on the servo horn aligns with the notch on the servo casing and that the bracket is also aligned as shown below:



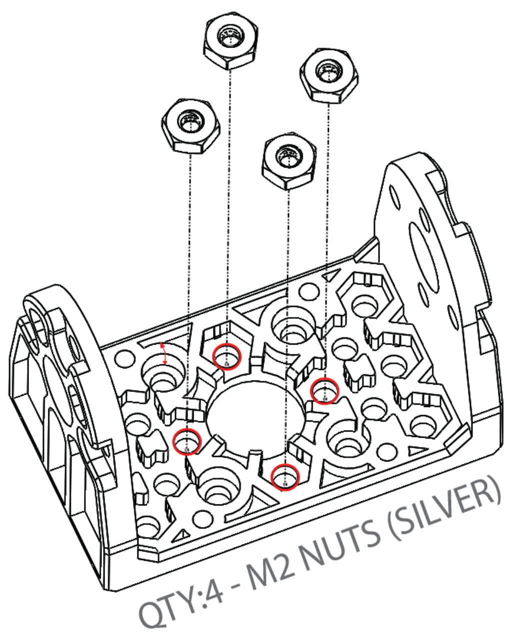
19. Attach the elbow actuator to XL-330_side_bracket using QTY:4 M2x8mm TAP screws.



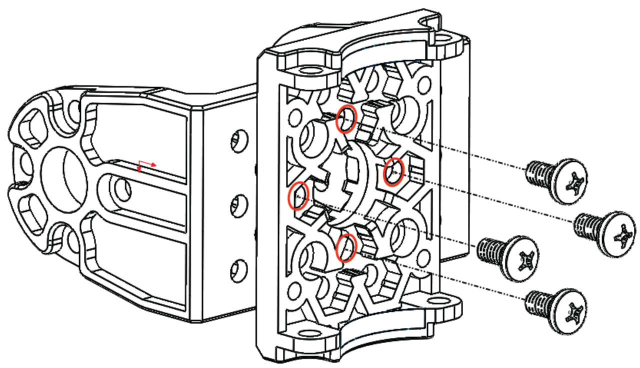
20. Attach XL-330_idle_bearing and XL-330_idle_horn to wrist actuator (XL330-M288-T, ID:4) using QTY:1 M2.6x6mm TAP screw.



21. Insert QTY:4 M2 nuts (silver colored) into the 4 closest holes surrounding the central hole in the FPX330-H101 wrist flexion bracket.

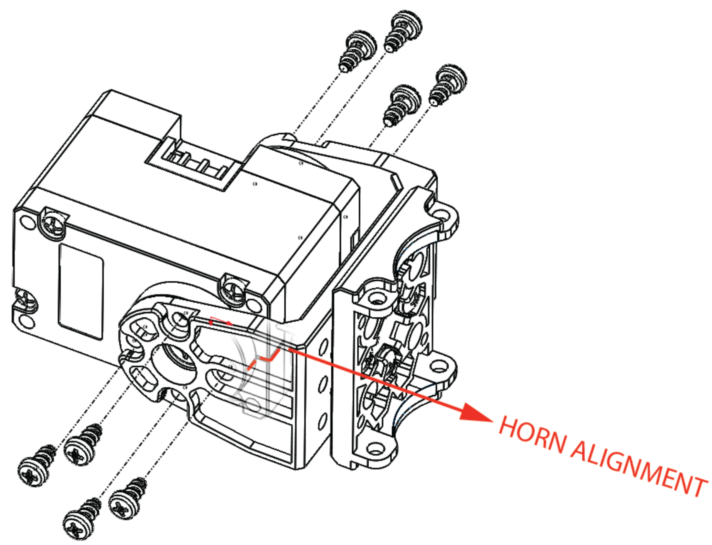


22. Attach the FPX330-S102 bottom bracket to the wrist flexion bracket using QTY:4 M2x4mm screws (silver colored).



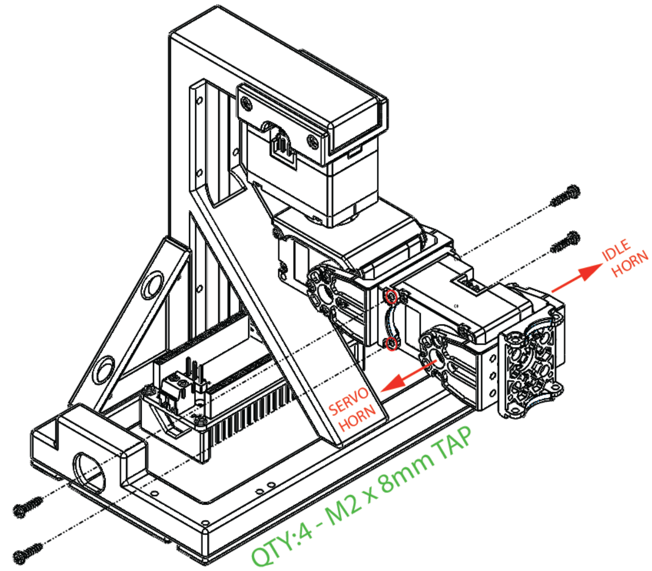
QTY:4 - M2 x 4mm (SILVER)

23. Attach the wrist flexion bracket to the wrist actuator using QTY:8 M2x4mm TAP screws. Ensure that the notch on the servo horn aligns with the notch on the servo casing and that the bracket is also aligned as shown below:

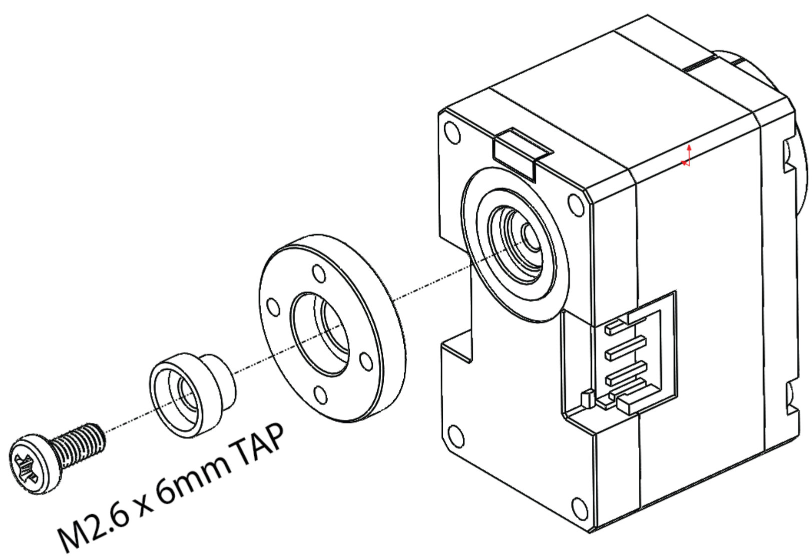


QTY:8 - M2 x 4mm TAP

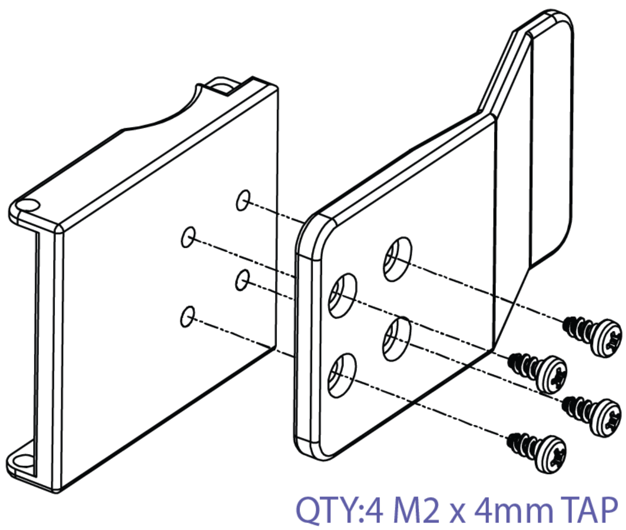
24. Attach the wrist actuator (XL330-M288-T, ID:4) to the FPX330-S102 bottom bracket using QTY:4 M2x8mm TAP screws.



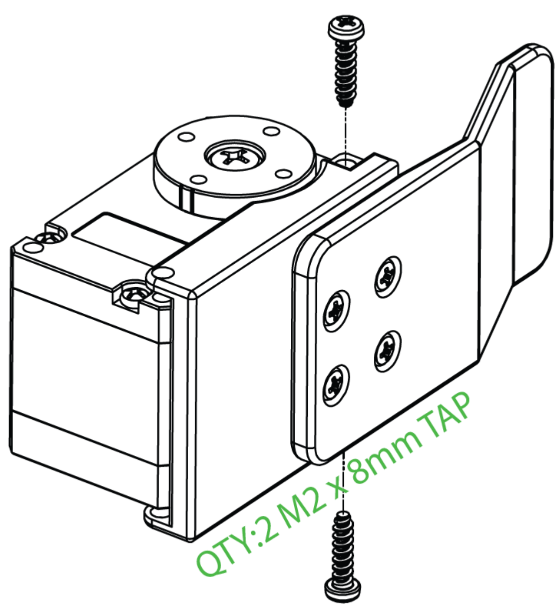
25. Attach XL-330_idle_bearing and XL-330_idle_horn to gripper actuator (XL330-M288-T, ID:5) using QTY:1 M2.6x6mm TAP screw.



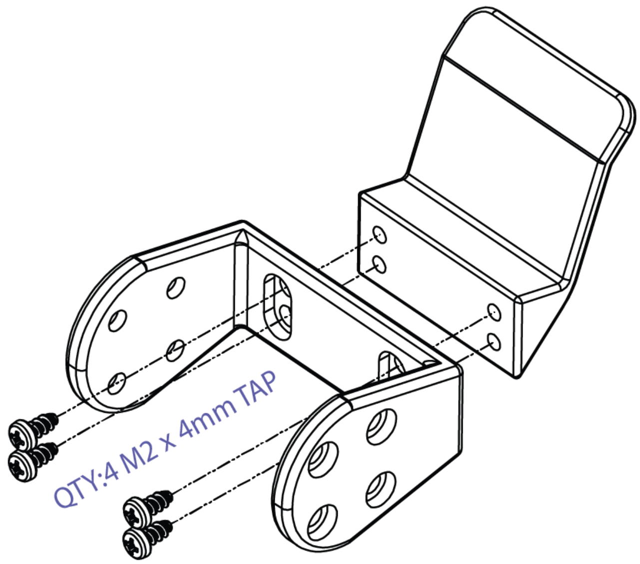
26. Attach the chopstick_fixed_plate to the chopsticks_side_bracket using QTY:4 M2x4mm TAP screws.



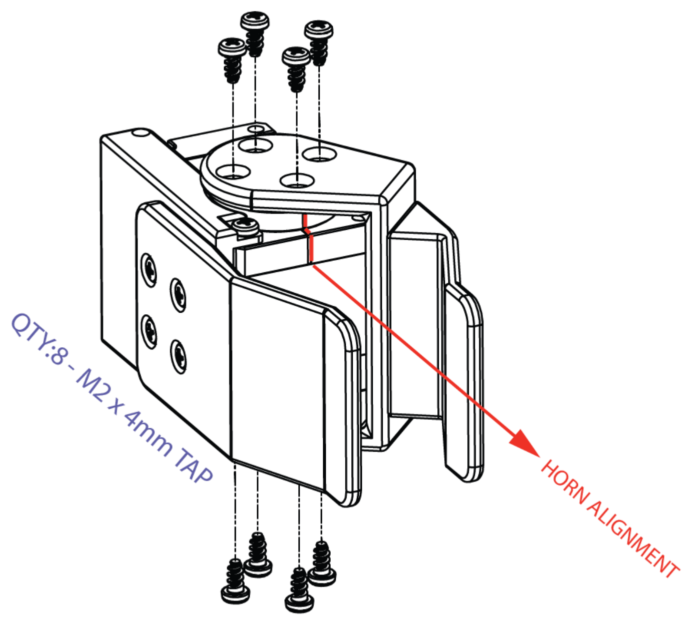
27. Attach the chopsticks_side_bracket to the gripper actuator (XL330-M288-T, ID:5) using QTY:2 M2x8mm TAP screws in the front of the actuator closest to the chopstick_fixed_plate



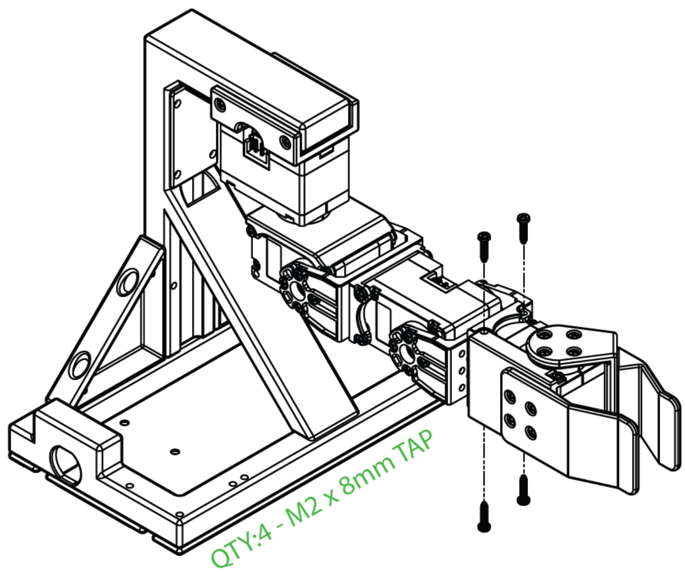
28. Attach chopsticks_moving_plate to chopsticks_flex_bracket using QTY:4 M2x4mm TAP screws.



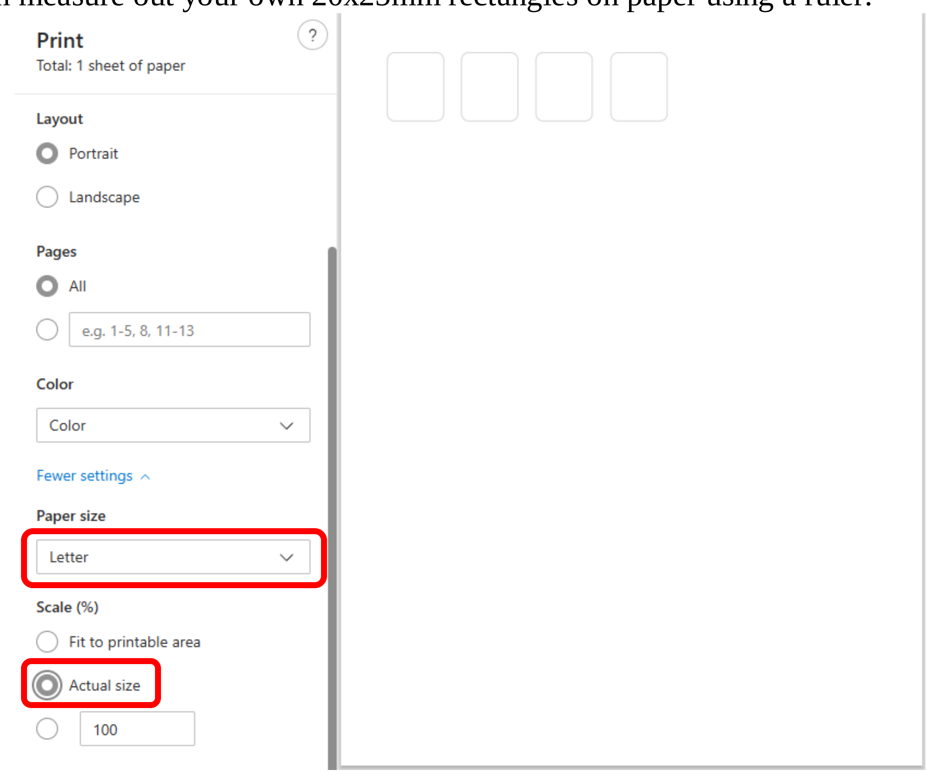
29. Attach chopsticks_flex_bracket to the hand actuator using QTY:8 M2x4mm TAP screws. Ensure that the notch on the servo horn aligns with the notch on the servo casing and that the bracket is also aligned as shown below:



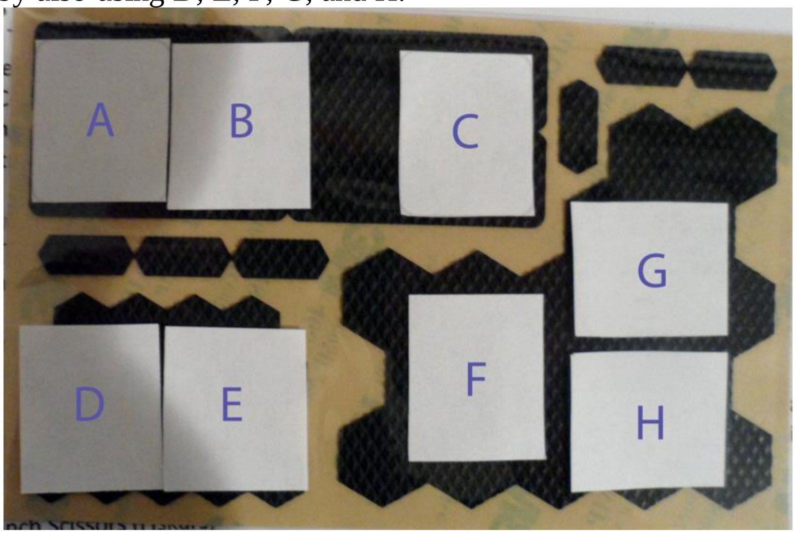
30. Attach the hand actuator assembly to the FPX330-S102 bottom bracket using QTY:4 M2x8mm TAP screws.



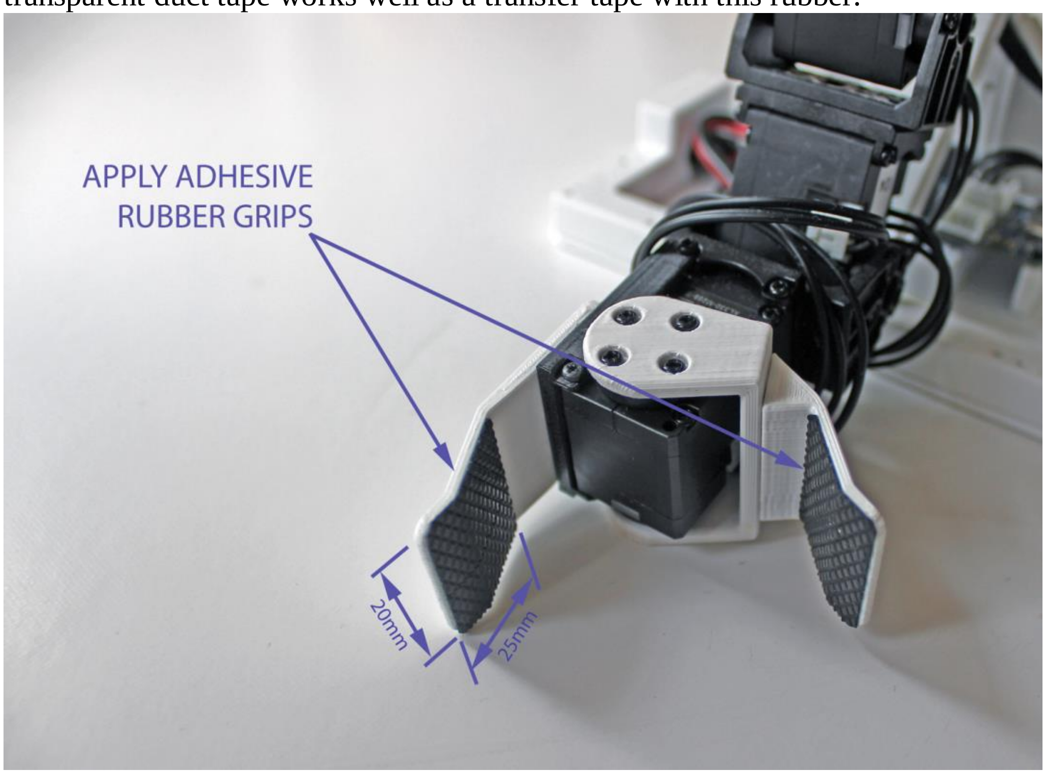
31. Print out the adhesive rubber gripper templates by opening "gripper_rubber_template_rev1.pdf" in a PDF viewer and setting the paper size to "letter" and a scale to "actual size". Alternatively, if you do not have a printer you can measure out your own 20x25mm rectangles on paper using a ruler.



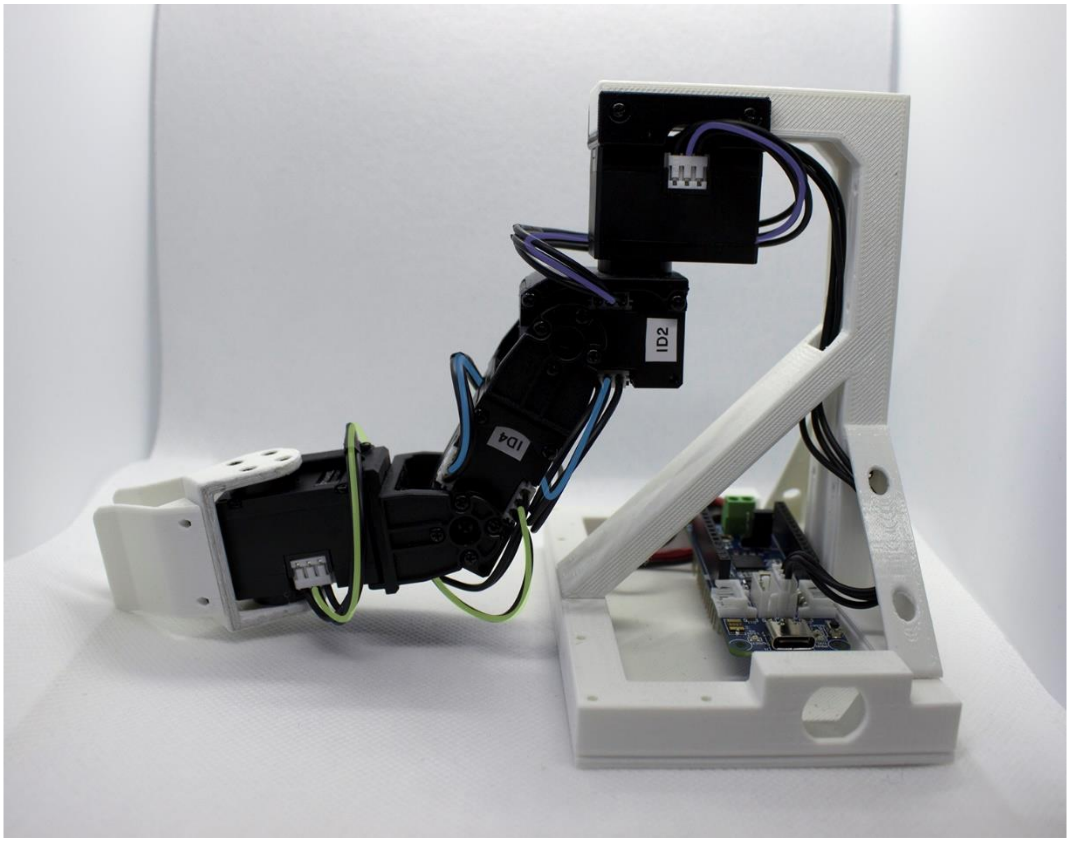
32. Cut out the paper templates using scissors and then place the paper templates over the adhesive rubber and cut out QTY:2 rectangular grips. If you are just making one kit then we recommend cutting out the rubber using positions A, B, or C as seen below. If you are making multiple kits then you can get more grips out of a single kit by also using D, E, F, G, and H.



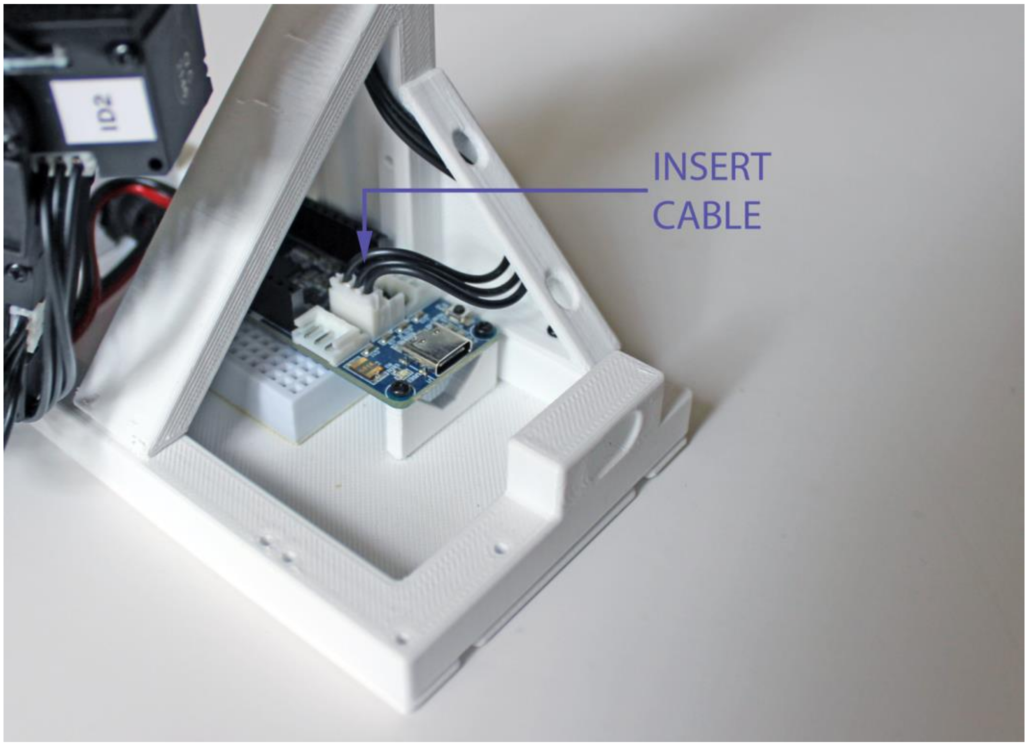
33. Apply the adhesive rubber grips to the Mini Bento Chopsticks gripper plates as seen in the following illustration. If you are using pieces from A, B, or C you can probably just place them by hand, but if you are using sections D, E, F, G, or H then we recommend using a transfer tape process (see video for more details) to avoid having to place all the small hexagons individually. We have found that transparent duct tape works well as a transfer tape with this rubber.



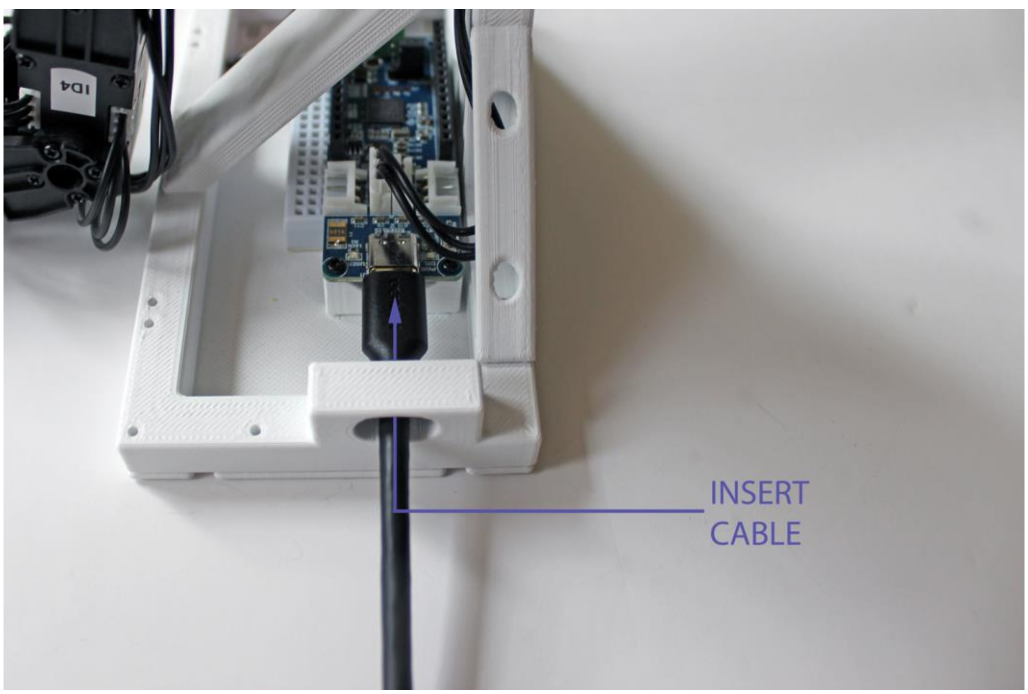
34. Connect the actuators together with the 180mm cables wrapping them around the servos as shown in the following image:
    a. Attach a 180mm cable from the shoulder actuator (ID:1) to the elbow actuator (ID:2).
    b. Attach a 180mm cable from the elbow actuator (ID:2) to the wrist actuator (ID:4)
    c. Attach a 180mm cable from the wrist actuator (ID:4) to the gripper actuator (ID:5)



35. Attach the 180mm cable from the shoulder actuator (ID:1) into one of the upward facing 3-pin connector ports on the OpenRB-150.

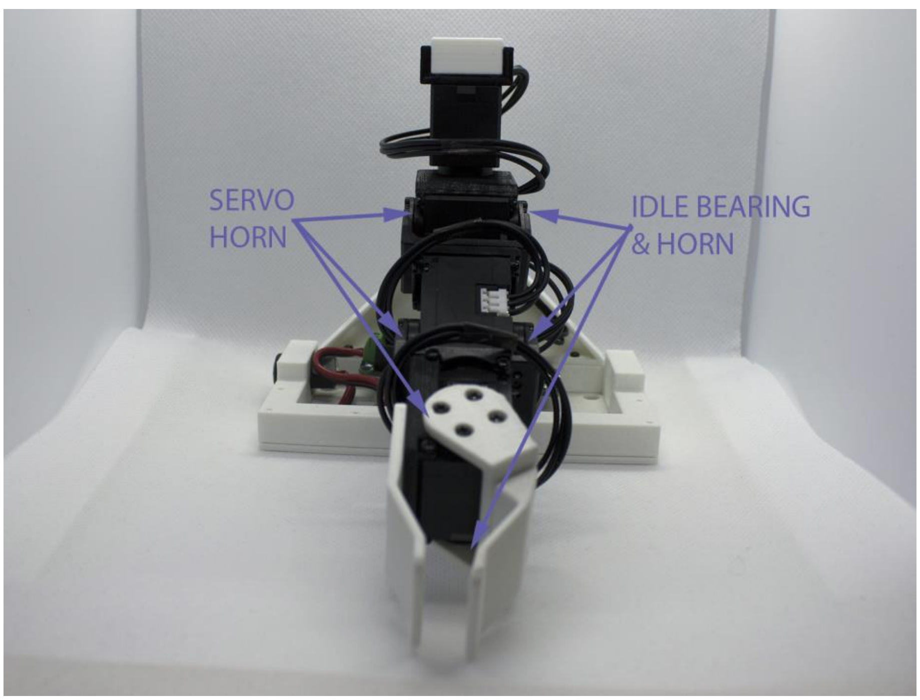


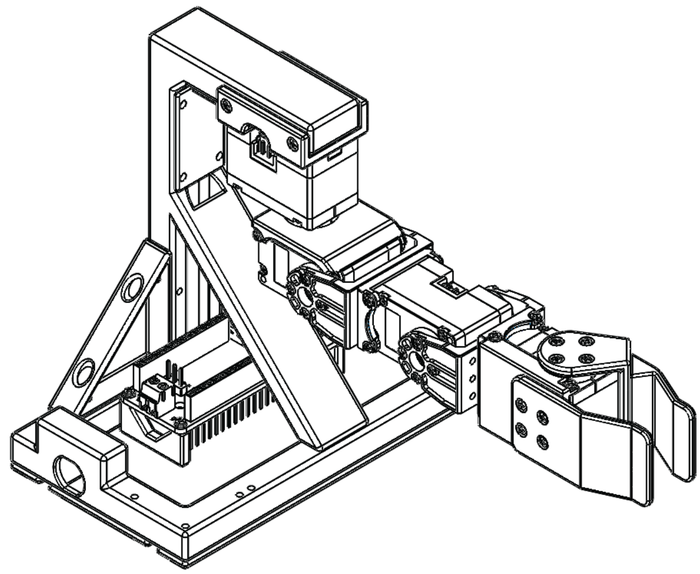
36. Insert USB C to USB A cable through the opening on the other side of stand_base and connect it to the USB C port on the OpenRB-150

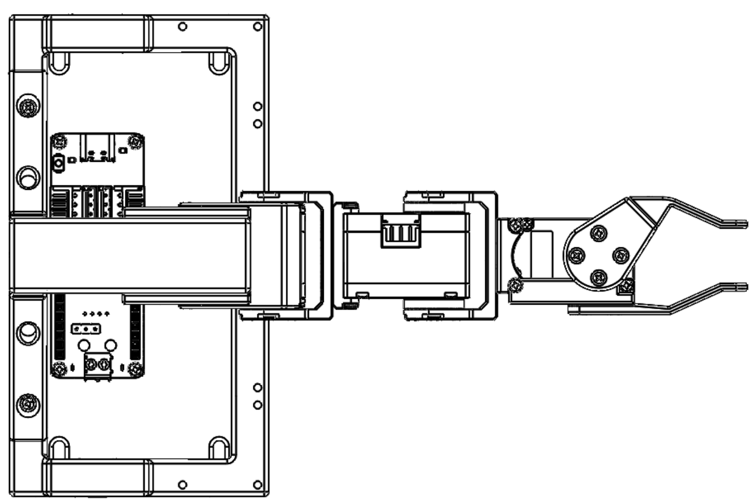


37. Connect the other end of the USB C to USB A cable to your computer.

38. Assembly of the arm is now complete! Here are some additional photos that may help with the assembly process:



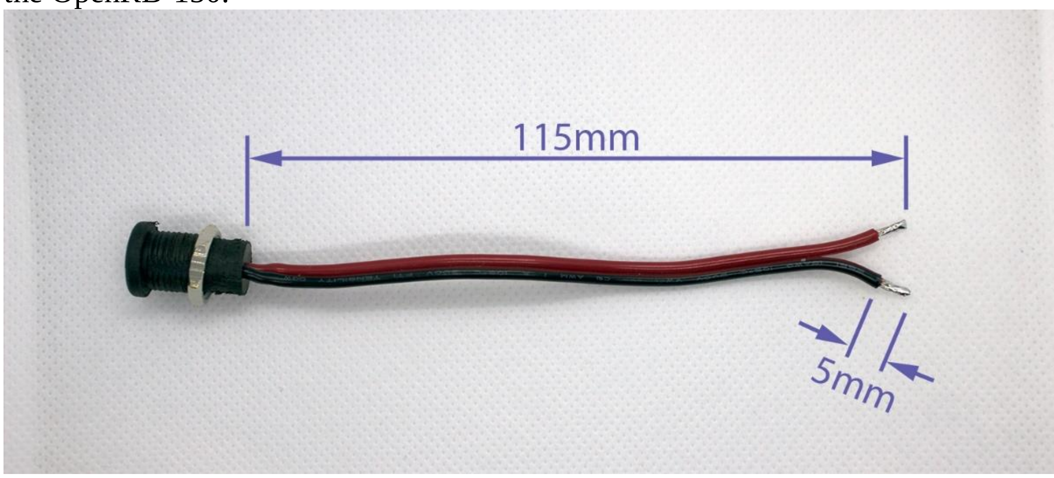




1.6 Optional Modifications

1. If the DC Barrel Jack (female) is still attached to the OpenRB-150 from the Dynamixel Servo programming, disconnect it by loosening the screw terminals using a screw driver and remove the wires.

2. Cut the wires on the DC Barrel Jack (female) to a length of 115mm using wire cutters. Separate the ends of wires, so that the separated length is about 15mm and then strip 5mm of insulation off the ends of each wire. Twist up the end of each wire to keep the strands together and then tin the ends of the wires using solder to help keep the strands together. If you do this step be sure to not add too much solder otherwise it may make it difficult to insert the wires into the terminals on the OpenRB-150.

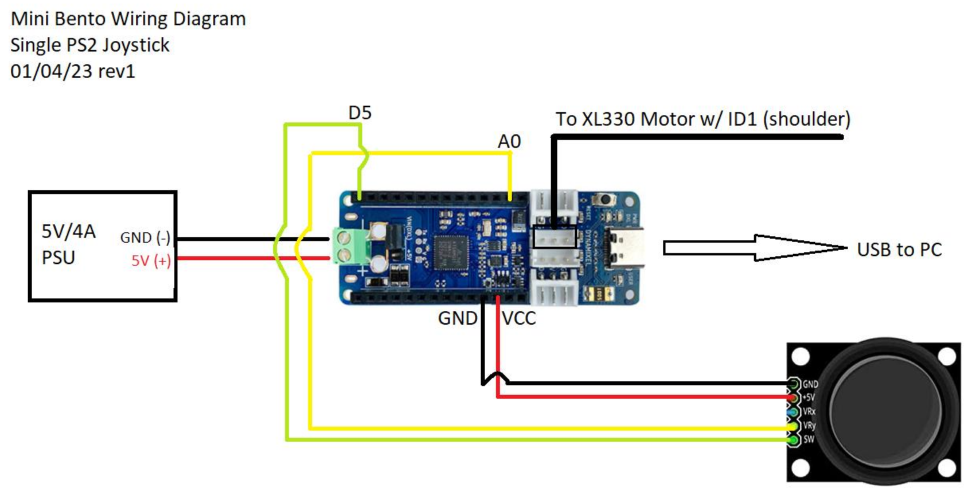


1.7 Arduino (Low Level Software)

Estimated Time: 30 minutes
Required Tools: Computer with Windows 10 or higher, OpenRB-150 or similar, 5V/4A power supply, DC Barrel Jack (female), USB C to USB A cable

1. Attach QTY:1 thumb joystick to the left side (when viewed from behind) of the stand_base using QTY:3 M2x8mm TAP screws. Ensure the header pins point towards the OpenRB-150.

2. With the power supply and USB cable unplugged, connect the prototyping wires between the thumb joystick and OpenRB-150 according to the following wiring diagram:



NOTE: It is possible to plug in the wires with everything assembled, but you can also temporarily remove the screws from the electronics plate and stand base or from the upper stand to make it easier before reassembling.

20. Power on the OpenRB-150 by plugging the 5V/4A power supply into a surge protector or wall outlet and connecting the barrel plug (male) of the power supply to the DC Barrel Jack (female). The green LED on the OpenRB-150 should illuminate indicating that the board has power. A few seconds later the red LED on the OpenRB-150 should also turn on indicating that the Dynamixel bus has initialized and is also now sending power to any connected Dynamixel servos.

21. Connect the OpenRB-150 to the computer using the USB C to USB A cable. If this is the first time you have connected the device Windows may give some notifications that a new device is being setup.

22. Download mini_bento_joint_velocity_control_rev2 sketch and extract it to a folder on your computer.

23. Open the “mini_bento_joint_velocity_control_rev2” sketch and upload it in the in the Arduino IDE
    a. Go “tools > board > OpenRB-150” and select “OpenRB-150”
    b. Go “tools > port” and select the option that has “(OpenRB-150)” next to it.
    c. Click “upload” in the “mini_bento_joint_velocity_control_rev2” sketch to upload the sketch to the OpenRB-150

24. The sketch will take a few seconds to upload and then you can control a single joint on the Mini Bento at a time by moving one of the axes on the joystick. Click the thumb joystick to switch sequentially through the joints (the LED of the selected actuator will glow red)

1.8 brachI/Oplexus (High Level Software)

Estimated Time: 30 min
Required Tools: Computer with Windows 10 or higher, OpenRB-150 or similar, 5V/4A power supply, DC Barrel Jack (female), USB C to USB A cable
NOTE: Support for the Mini Bento Arm in brachI/Oplexus is in beta. If you would like to try it out please contact us and we can send you early access to the project files.

1. Download the “usb_to_dynamixel” sketch and extract it to a folder on your computer.

2. Open the “usb_to_dynamixel” sketch and upload it in the in the Arduino IDE
    a. Go “tools > board > OpenRB-150” and select “OpenRB-150”
    b. Go “tools > port” and select the option that has “(OpenRB-150)” next to it.
    c. Go “file > examples > OpenRB-150” and select the “usb_to_dynamixel” sketch. A new window should open in the Arduino IDE that is titled “usb_to_dynamixel”
    d. Click “upload” in the “usb_to_dynamixel” sketch to upload the sketch to the OpenRB-150

3. If you haven’t already as part of installing the Dynamixel Wizard 2.0, download and install the Microsoft Visual C++ 2015-2019 Redistributable (x86) from the following link:

    https://www.dropbox.com/s/3e5horhq73hm677/VC_redist.x86.exe?dl=0

    a. If the redistributable is already installed on your computer then you don’t need to reinstall it. You can check to see if it is installed by searching for “add or remove programs” and then looking to see if “Microsoft Visual C++ 2015-2019 Redistributable (x86) is present.

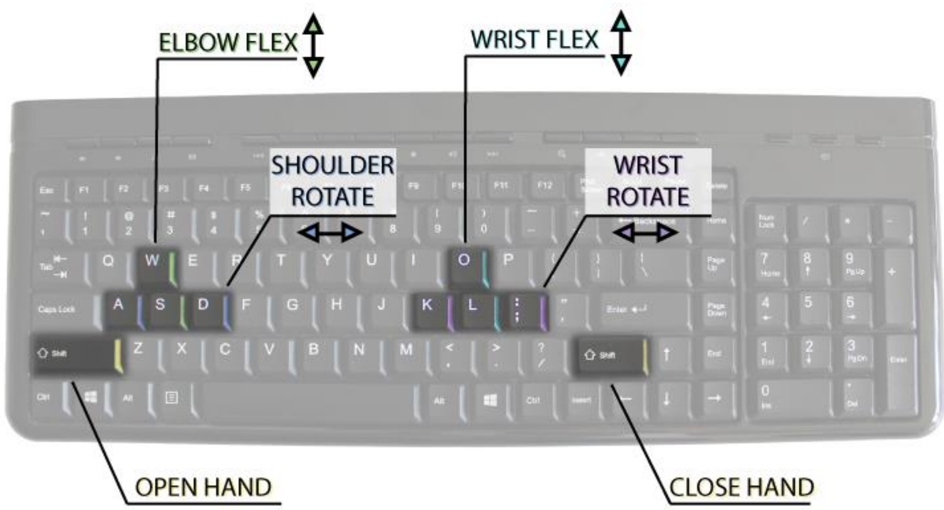
4. [Contact us](#) to request the special version of the brachI/Oplexus software that works with the Mini Bento Arm and extract it to a folder on your desktop.

5. Navigate to the path with the executable:

   ```
   GUI_DOF6_12ch_BENTO_rev52b_dyna2b\Example1\bin\x86\Debug
   ```

6. Click on "brachIOplexus.exe" to run brachI/Oplexus
   a. You can also right click on the executable and select "create shortcut" and then drag onto your desktop for easy access

7. Once brachI/Oplexus opens you can try out the Virtual Bento Arm by clicking the "Launch Virtual Bento Demo" in the "Input/Output" tab
   a. Here is the mapping to control the Virtual Bento Arm with the keyboard



8. Once you have finished trying the Virtual Bento Arm you can close the unity window and go back to the input/output tab and click "connect" in the "Bento Arm – Setup" groupbox
9. Now you can try controlling your Mini Bento Arm with the keyboard by navigating to the "Bento Arm" tab in brachI/Oplexus and clicking "Torque On" and "Run"